# DS35X4GBXXX

## 3.3V/1.8V x1/x2/x4 4G-bit SPI NAND FLASH

**Revision History** ..... 5
**FEATURES** ..... 6
**Part Numbering System** ..... 7
1 SUMMARY DESCRIPTION ..... 8
1.1 Product List ..... 9
1.2 Pin description ..... 10
1.3 Functional block diagram ..... 11
1.4 Address Map ..... 11
1.5 Command Set ..... 12
2 BUS OPERATION ..... 13
2.1 SPI Mode. ..... 13
2.2 CS# ..... 13
2.3 SI/SIO0 ..... 13
2.4 SO/SIO1 ..... 13
2.5 WP#/SIO2 ..... 13
2.6 HOLD#/SIO3 ..... 14
3 DEVICE OPERATION ..... 15
3.1 Write Operations. ..... 15
3.2 Feature Operations ..... 16
3.3 Read Operations. ..... 17
3.4 Read ID. ..... 22
3.5 Parameter Page. ..... 23
3.6 UniqueID Page. ..... 24
3.7 Program Operations. ..... 24
3.8 BLOCK ERASE ..... 30
3.9 Block Lock Feature. ..... 31
3.10 OTP Feature. ..... 32
3.11 Status Register ..... 33
3.12 ECC Protection ..... 34
3.13 Block Protection (One Time Program) ..... 35
4 Device Parameters ..... 36
5 Bad Block Management ..... 41
6 Supported Packages ..... 43

## Figure Content

**Figure 1.1** Logic Diagram ..... 9
**Figure 1.2** Block Description ..... 11
**Figure 1.3** Address Map ..... 11
**Figure 2.1** SPI Modes Timing ..... 13
**Figure 3.1** Write Enable ..... 15
**Figure 3.2** Write Disable ..... 15
**Figure 3.3** Get Features(0Fh) ..... 16
**Figure 3.4** Set Features(1Fh) ..... 16
**Figure 3.5** Page Read (13h) ..... 18
**Figure 3.6** Random Data Read (03h or 0Bh) ..... 19
**Figure 3.7** Read From Cache x2 ..... 20
**Figure 3.8** Read From Cache x4 ..... 21
**Figure 3.9** Read ID (9Fh) ..... 22
**Figure 3.10** Program Load (02h) ..... 25
**Figure 3.11** Program Load x4 (32h) ..... 26
**Figure 3.12** Program Excute (10h) ..... 27
**Figure 3.13** Program Load Random Data (84h) ..... 28
**Figure 3.14** Program Load Random Data x4 (34h) ..... 29
**Figure 3.15** Block Erase ..... 30
**Figure 4.1** Serial Input Timing ..... 39
**Figure 4.2** Serial Output Timing ..... 39
**Figure 4.3** Hold Timing ..... 39
**Figure 4.4** WP# Setup/Hold Timing When BPRWD=1 ..... 40
**Figure 4.5** Power on Sequence ..... 40
**Figure 5.1** Bad Block Management Flowchart ..... 41
**Figure 5.2** Bad Block Replacement ..... 42
**Figure 6.1** Pin Configuration ..... 43
**Figure 6.2** WSON (8*6mm) ..... 44
**Figure 6.3** SOP16 ..... 45
**Figure 6.4** BGA24(5x5 Ball Array) ..... 46

## Table Content

**Table 1.1** Signal Name....9
**Table 1.2** Pin Description....10
**Table 1.3** Command Set....12
**Table 3.1** Status Register Coding....16
**Table 3.2** Read ID for Supported Configurations....22
**Table 3.3** Parameter Page Data....24
**Table 3.4** Definition of Protectiong Bits....31
**Table 3.5** OTP States....32
**Table 3.6** Status Register Bit Descriptions....33
**Table 3.7** The Distribution of ECC Segment and Spare Area....34
**Table 4.1** Valid Blocks Number....36
**Table 4.2** Absolute Maximum Ratings....36
**Table 4.3** DC and Operating Characteristics....36
**Table 4.4** AC Test Conditions....37
**Table 4.5** Ping Capacitance (TA=25C,f=1.0MHz)....37
**Table 4.6** Read/Program/Erase Characteristics....37
**Table 4.7** AC Timing Characteristics....38
**Table 5.1** Block Failure....42

## Documents title

4Gbit NAND FLASH

## Revision History

| Revision No. | History | Draft date | Release date | Remark |
|---|---|---|---|---|
| 0.0 | Initial draft | Jul.10.2023 | Jul.18.2023 | |
| 0.1 | Delete 12$^{th}$ bit plane select requirement | Oct.06.2023 | Oct.09.2023 | |
| 0.2 | Update protected block range | Nov.10.2023 | Nov.14.2023 | |
| 0.3 | Delete internal data move within lower/higher 2048 blocks limitation description | May.20.2024 | May.23.2024 | |
| 0.4 | Add automotive grade2 product info | Sep.20.2024 | Sep.22.2024 | |
| 0.5 | Update P/E cycling to 80K | Nov.26.2024 | Nov.27.2024 | |
| 0.6 | Add BGA24 package info | Mar.03.2025 | Mar.06.2025 | |
| 0.7 | Add 125°C product P/N support | Mar.12.2025 | Mar.14.2025 | |

***Datasheet can be modified without any notice!***

# FEATURES

## ■ Serial Peripheral Interface

- Mode 0 and Mode 3

## ■ Standard, Dual, Quad SPI

- Standard SPI: SCLK, CS#, SI, SO
- Dual SPI: SCLK, CS#, SIO0, SIO1
- Quad SPI: SCLK, CS#, SIO0, SIO1, SIO2, SIO3

## ■ SUPPLY VOLTAGE

- VCC = 1.8/3.3 Volt core supply voltage for Program, Erase and Read operations

## ■ PAGE READ / PROGRAM

- (2048+128 spare) byte
- Random access: 25us (w/o ECC), 130us(w/ ECC)
- Serial access: 83MHz (1.8V) 104MHz (3.3V)
- Page program time: 300us (Typ)

## ■ FAST BLOCK ERASE

- Block size: (128K + 8K) bytes
- Block erase time: 2ms (Typ)

## ■ MEMORY CELL ARRAY

- (2K + 128) bytes x 64 pages x 4096 blocks

## ■ ELECTRONIC SIGNATURE

- Manufacturer Code
- Device Code

## ■ STATUS REGISTER

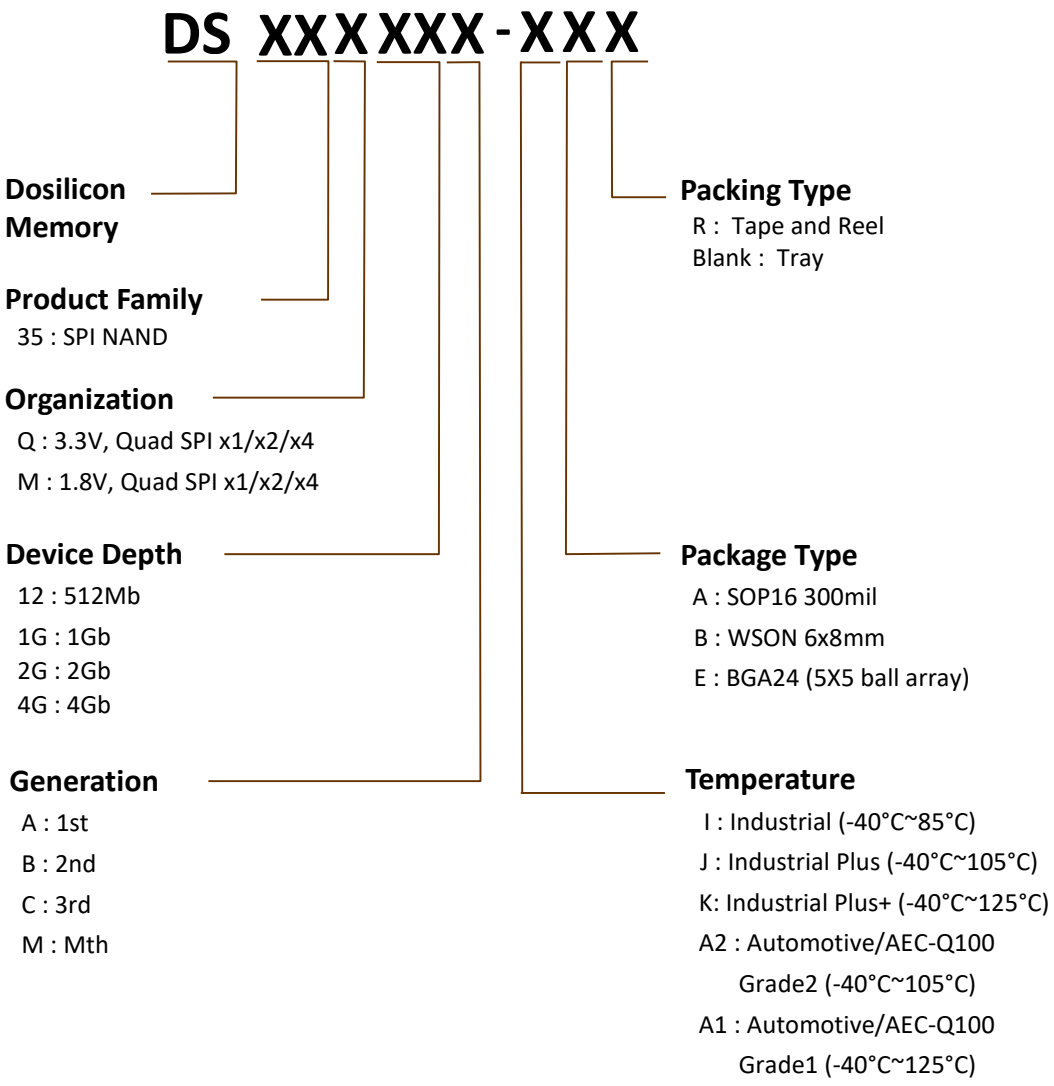
## ■ HARDWARE DATA PROTECTION

- Enable/Disable protection with WP# Pin
- Top or Bottom, Block selection combination

## ■ DATA RETENTION

- Max cycling: 80K Program / Erase cycles
- Data retention: 10 Years (8bit/512byte ECC)
- Internal ECC can be enabled (8bit ECC)
- Block zero is a valid block and will be valid for at least 1K program-erase cycles with ECC

## Part Numbering System



# 1 SUMMARY DESCRIPTION

**DS35X4GBXXX** is a **512Mx8bit** with spare **32Mx8** bit capacity.
The device is offered in 3.3/1.8 Vcc Power Supply, and with SPI interface.
The memory is divided into blocks that can be erased independently so it is possible to preserve valid data while old data is erased.
The device contains **4096 blocks**, composed by 64 pages consisting in two NAND structures of 32 series connected Flash cells. Program operation allows the 2176-byte page writing in typical 300us and an erase operation can be performed in typical 2 ms on a 128K-byte block.
Data in the page can be read out at **10ns** cycle time per word **(3.3V version)**, and at **12ns** cycle time per word **(1.8V version)**. The on-chip Program/Erase Controller automates all program and erase functions including pulse repetition, where required, and internal verification and margining of data.

An internal 8-bit ECC logic is implemented in the chip, which is enabled by default. The internal ECC can be disabled or enabled again by command. When the internal 8-bit ECC logic is disabled, the host side needs to handle the 8-bit ECC by host micro controller.

The serial peripheral interface (SPI) provides NAND Flash with a cost-effective non-volatile memory storage solution in systems where pin count must be kept to a minimum. It is also an alternative to SPI-NOR, offering superior write performance and cost per bit over SPI-NOR.

## 1.1 Product List

| PART NUMBER | ORGANIZATION | VCC RANGE | PACKAGE |
|---|---|---|---|
| DS35M4GB | X1/X2/X4 | 1.7 – 1.95 Volt | 16SOP, 8WSON, BGA24 |
| DS35Q4GB | X1/X2/X4 | 2.7 – 3.6 Volt | 16SOP, 8WSON, BGA24 |

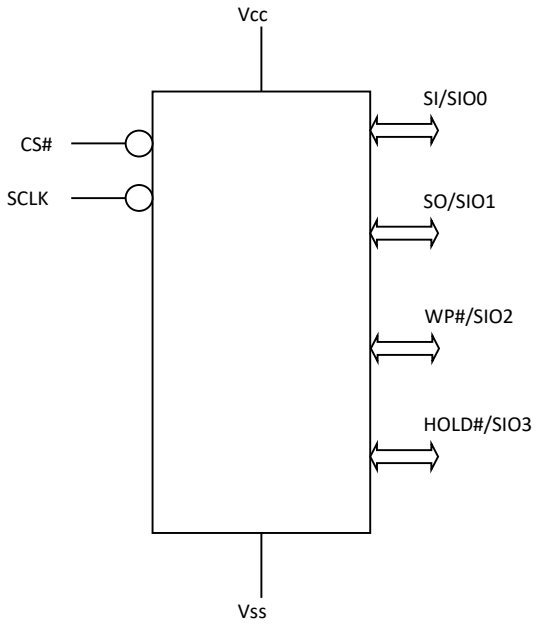



**Figure 1.1** Logic Diagram

| Name | Function |
|---|---|
| CS# | Chip Select |
| SCLK | Clock Input |
| SI/SIO0 | Serial Data Input (for 1 x I/O) or<br>Serial Data Input & Output (for 2xI/O or 4xI/O mode) |
| SO/SIO1 | Serial Data Output (for 1 x I/O) or<br>Serial Data Input & Output (for 2xI/O or 4xI/O mode) |
| WP#/SIO2 | WP# or<br>Serial Data Input & Output (for 4xI/O mode) |
| HOLD#/SIO3 | Hold# or<br>Serial Data Input & Output (for 4xI/O mode) |
| Vcc | Power supply |
| Vss | Ground |
| NC | No Connection |

**Table 1.1** Signal Name

## 1.2 Pin description

| Pin Name | Description |
|---|---|
| **CS#** | **Chip select**<br>Places the device in active power mode when driven LOW. Deselects the device and places SO at High-Z when HIGH |
| **SCLK** | **Serial clock**<br>Provides serial interface timing. Latches commands, addresses, and data on SI on the rising edge of SCLK. Triggers output on SO after the falling edge of SCLK. |
| **SI/SIO0** | **Serial data input**<br>Transfers data serially into the device. Device latches commands, addresses, and program data on SI on the rising edge of SCLK. SI must not be driven by the host during x2 or x4 read operations. |
| **SO/SIO1** | **Serial data output**<br>Transfers data serially out of the device on the falling edge of SCLK. |
| **WP#/SIO2** | **Write protect**<br>When LOW, prevents overwriting block-lock bits if the block register write disable (BRWD) bit is set.<br>WP# must not be driven by the host during x4 read operations. |
| **HOLD#/SIO3** | **Hold**<br>Pauses any serial communication with the device without deselecting it. When driven LOW, SO is at High-Z, and all inputs at SI and SCLK are ignored. Requires that CS# also be driven LOW.<br>HOLD# must not be driven by the host during x4 read operations. |
| **$V_{CC}$** | **Supply Voltage**<br>The VCC supplies the power for all the operations (Read, Write, Erase). |
| **$V_{SS}$** | **Ground** |
| **NC / DNU** | **No Connection / Don't Use** |

**Table 1.2** Pin Description

## 1.3 Functional block diagram

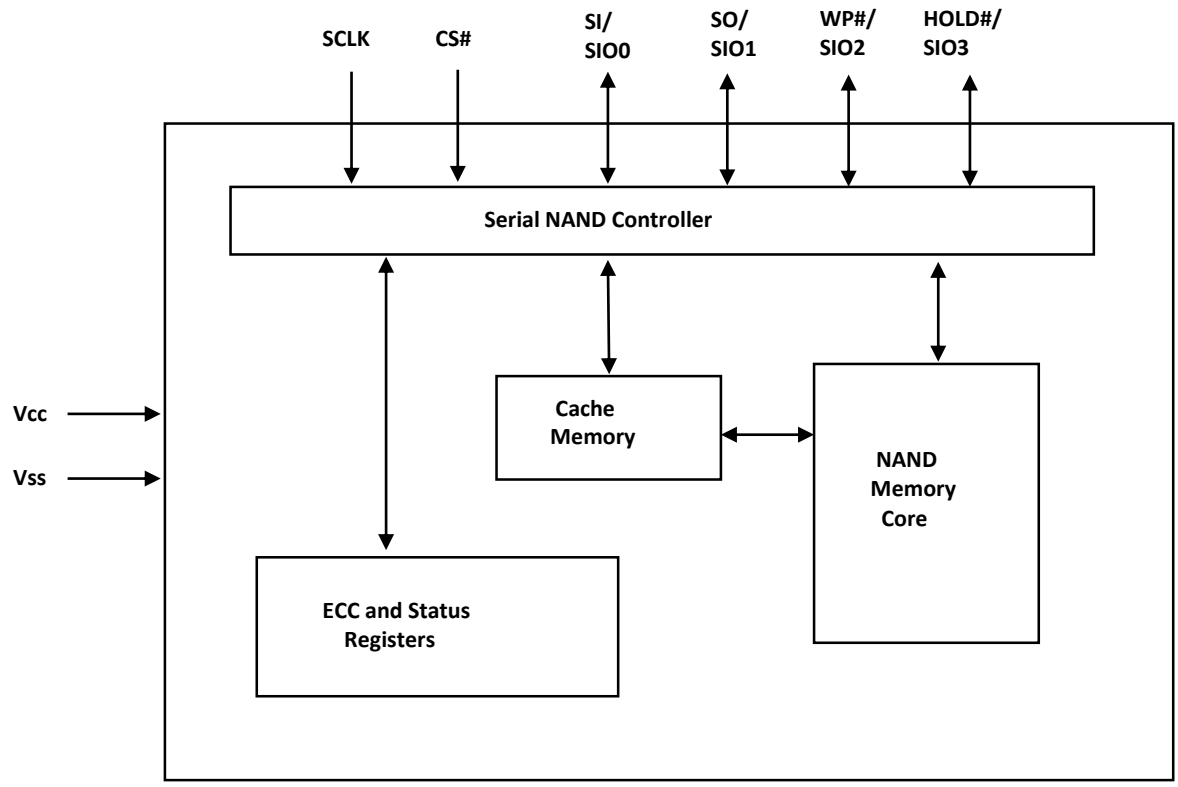



**Figure 1.2** Block Description

## 1.4 Address Map

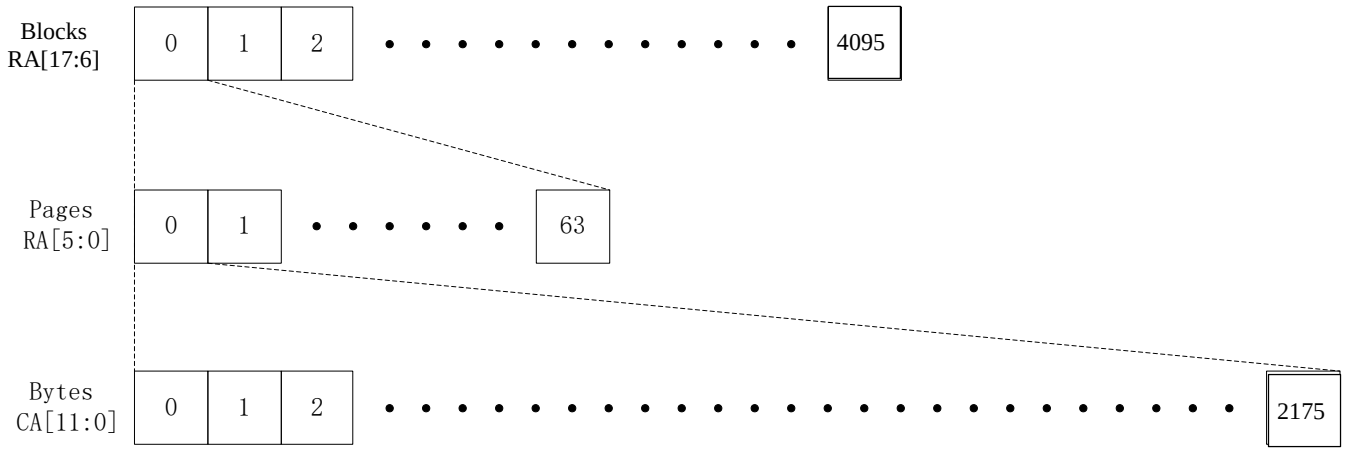



**Figure 1.3** Address Map

## 1.5 Command Set

| FUNCTION | Command Code | Address Bytes | Dummy Bytes | Data Bytes | Comments |
|---|---|---|---|---|---|
| GET FEATURE | 0Fh | 1 | 0 | 1 | Get features |
| SET FEATURE | 1Fh | 1 | 0 | 1 | Set features |
| WRITE ENABLE | 06h | 0 | 0 | 0 | |
| WRITE DISABLE | 04h | 0 | 0 | 0 | |
| PAGE READ | 13h | 3 | 0 | 0 | Array read |
| READ FROM CACHE | 03h, 0Bh | 2 | 1 | 1 to 2176 | Output cache data on SO |
| READ FROM CACHE x2 | 3Bh | 2 | 1 | 1 to 2176 | Output cache data on SI and SO |
| READ FROM CACHE x4 | 6Bh | 2 | 1 | 1 to 2176 | Output cache data on SI, SO, WP#,HOLD# |
| PROGRAM LOAD | 02h | 2 | 0 | 1 to 2176 | Load program data with cache reset |
| PROGRAM LOAD x4 | 32h | 2 | 0 | 1 to 2176 | Load program data on SI, SO, WP#,HOLD# with cache reset |
| PROGRAM LOAD RANDOM DATA | 84h | 2 | 0 | 1 to 2176 | Load program data without cache reset |
| PROGRAM LOAD RANDOM DATA x4 | 34h | 2 | 0 | 1 to 2176 | Load program data on SI, SO, WP#,HOLD# without cache reset |
| PROGRAM EXECUTE | 10h | 3 | 0 | 0 | Enter block/page address, no data, execute |
| BLOCK ERASE | D8h | 3 | 0 | 0 | Block erase |
| READ ID | 9Fh | 0 | 1 | 2 | Read device ID |
| RESET | FFh | 0 | 0 | 0 | Reset the device |
| BLOCK PROTECTION 1 | B1h | 3 | 0 | 0 | Block protection range 1 |
| BLOCK PROTECTION 2 | B2h | 3 | 0 | 0 | Block protection range 2 |
| BLOCK PROTECTION 3 | B3h | 3 | 0 | 0 | Block protection range 3 |
| BLOCK PROTECTION 4 | B4h | 3 | 0 | 0 | Block protection range 4 |

**Table 1.3** Command Set

# 2 BUS OPERATION

## 2.1 SPI Mode.

Two SPI modes are supported.
• CPOL = 0, CPHA = 0 (Mode 0)
• CPOL = 1, CPHA = 1 (Mode 3)
Input data is latched in on the rising edge of SCLK, and output data is available from the falling edge of SCLK for both modes.
When the bus master is in standby mode:
• SCLK remains at 0 for (CPOL = 0, CPHA = 0 Mode 0)
• SCLK remains at 1 for (CPOL = 1, CPHA = 1 Mode 3)
See **Figure 2.1**

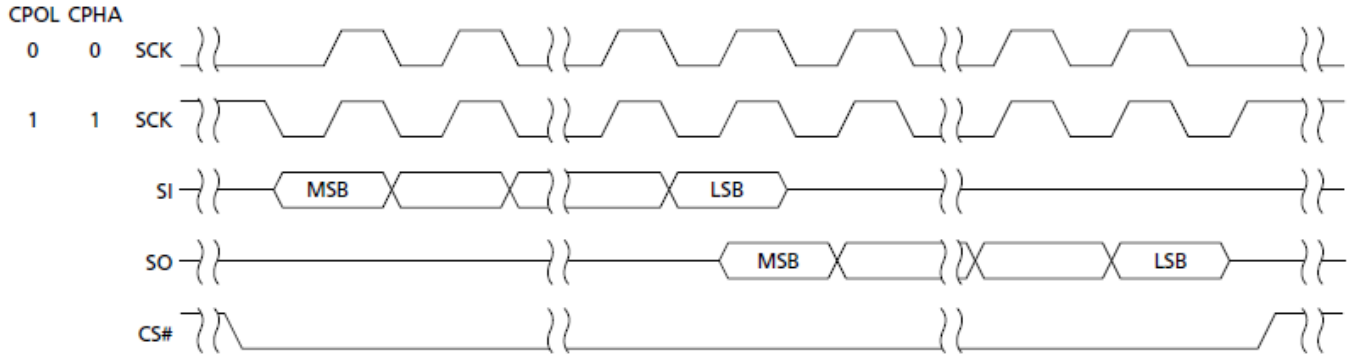



**Figure 2.1** SPI Modes Timing

## 2.2 CS#

Chip select (CS#) activates or deactivates the device. When CS goes LOW, the device is placed in active mode. When CS is HIGH, the device is placed in inactive mode and SO is High-Z.

## 2.3 SI/SIO0

Writes use serial data in (SI). Data, commands, and addresses are transferred on SI in x1 mode at the rising edge of SCLK. SI must not be driven by the host during x2 or x4 read operations.
SIO0 operation is enabled by issuing a READ FROM CACHE x2 or x4 command with data being clocked out of the device at the falling edge of SCLK. During this time the host must wait until the READ FROM CACHE x2 or x4 command is complete before driving SI.

## 2.4 SO/SIO1

Reads use serial data out (SO). Device reads are performed in x1, or x2, or x4 modes. SO acts as the only output in x1 READ operations, and as SIO1 in x2 and x4 read operations.
Data is clocked out of the device on SO at the falling edge of SCLK control signals.

## 2.5 WP#/SIO2

Write protect (WP#) prevents the block lock bits (BP0, BP1, and BP2) from being overwritten.

If the BRWD bit is set to 1 and WP# is LOW, the block protect bits cannot be altered. WP# must not be driven by the host during READ FROM CACHE x4 operations.
SIO2 operation is enabled by issuing a READ FROM CACHE x4 command with data being clocked out of the device at the falling edge of SCLK. During this time the host must wait until the READ FROM CACHE x4 command is complete before driving WP#.

### 2.6 HOLD#/SIO3

HOLD# input provides a method to pause serial communication with the device but does not terminate any ERASE, READ, or WRITE operation currently in progress.
Hold mode starts at the falling edge of HOLD# provided SCLK is also LOW. If SCLK is HIGH when HOLD# goes LOW, hold mode begins after the next falling edge of SCLK. Similarly, hold mode is exited at the rising edge of HOLD# provided SCLK is also LOW. If SCLK is HIGH, hold mode ends after the next falling edge of SCLK.
During hold mode, SO is High-Z, and SI and SCLK inputs are ignored.
SIO3 operation is enabled by issuing a READ FROM CACHE x4 command with data being clocked out of the device at the falling edge of SCLK. During this time the host must wait until the READ FROM CACHE x4 command is complete before driving HOLD#.

# 3 DEVICE OPERATION

## 3.1 Write Operations.

The WRITE ENABLE (06h) command sets the WEL bit in the status register to 1. WRITE ENABLE is required in the following operations that change the contents of the memory array:

- Page program
- OTP program
- BLOCK ERASE

See **Figure 3.1**

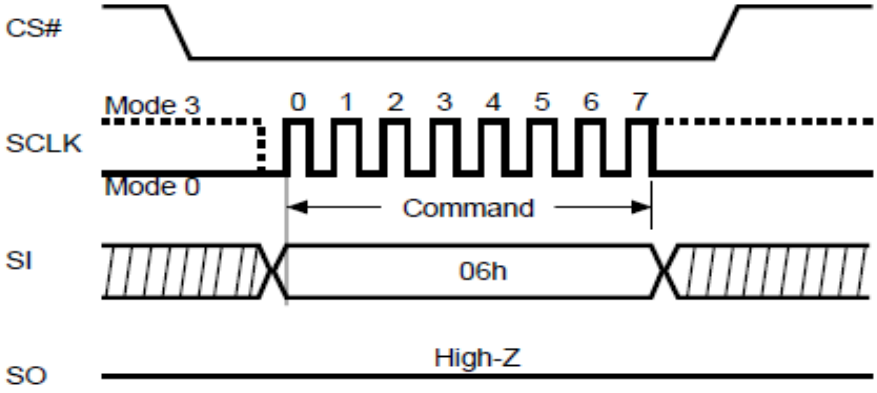



**Figure 3.1** Write Enable

The WRITE DISABLE (04h) command clears the WEL bit in the status register to 0. This disables the following operations:

- Page program
- OTP program
- BLOCK ERASE

See **Figure 3.2**

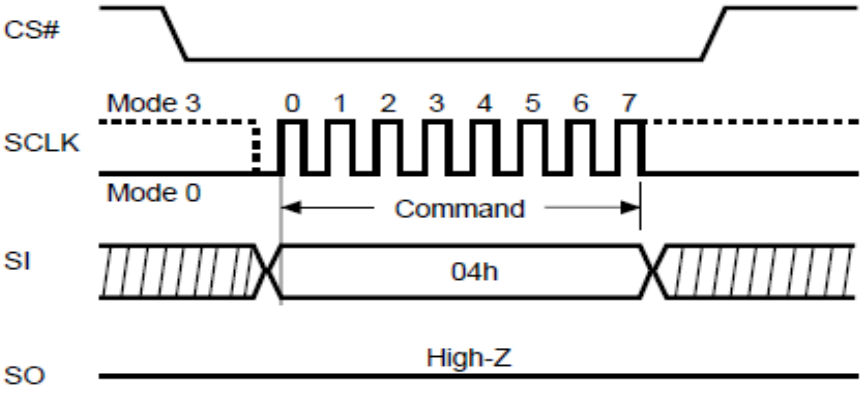



**Figure 3.2** Write Disable

### 3.2 Feature Operations

The GET FEATURES (0Fh) and SET FEATURES (1Fh) commands are used to alter the device behavior from the default power-on behavior. These commands use a 1-byte feature address to determine which feature is to be read or modified. Features such as OTP and block locking can be enabled or disabled by setting specific bits in feature address A0h and B0h (shown in the following table). The status register is mostly read, except WEL, which is a writable bit with the WRITE ENABLE (06h) command.
When a feature is set, it remains active until the device is power cycled or the feature is written to. Unless specified in the following table, once the device is set, it remains set, even if a RESET (FFh) command is issued. See **Figure 3.3, 3.4.**

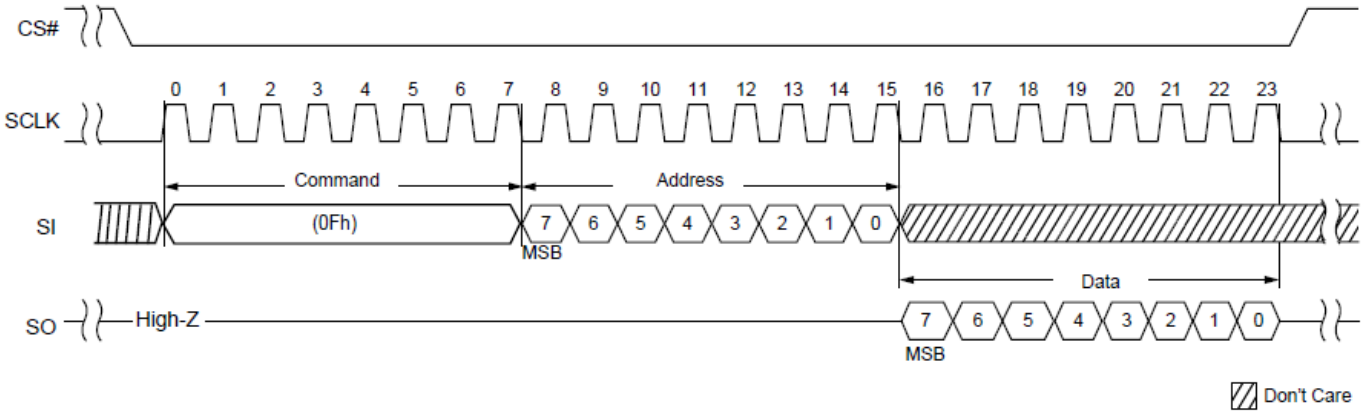



**Figure 3.3** Get Features(0Fh)

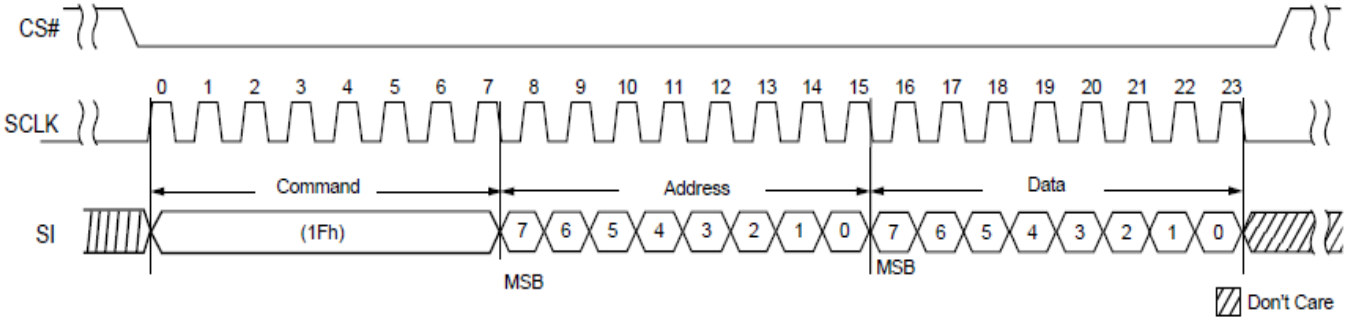



**Figure 3.4** Set Features(1Fh)

| Register | Address | Data Bits | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Block lock | A0h | BRWD | Reserved | BP2 | BP1 | BP0 | INV | CMP | Reserved |
| OTP | B0h | OTP_PRT | OTP_EN | Reserved | ECC Enable | Reserved | Reserved | Reserved | QE |
| Status | C0h | Reserved | ECC_S2 | ECC_S1 | ECC_S0 | P_Fail | E_Fail | WEL | OIP |
| Driver Strength | D0h | Reserved | DS_IO1 | DS_IO0 | Reserved | Reserved | Reserved | Reserved | Reserved |

**Table 3.1** Status Register Coding

### 3.3 Read Operations.

The device supports "Power-on Read" function, after power up, the device will automatically load the data of the 1st page of 1st block from array to cache. The host micro-controller may directly read the 1st page of 1st block data from the cache buffer. The data is also under the internal ECC protection.

The PAGE READ (13h) command transfers the data from the NAND Flash array to the cache register. The command sequence is follows:

- 13h (PAGE READ to cache)
- 0Fh (GET FEATURES command to read the status)
- 0Bh / 03h / 3Bh / 6Bh (Random data read)

The PAGE READ command requires a 24-bit address consisting of 6 dummy bits followed by a 18-bit block/page address. After the block/page addresses are registered, the device starts the transfer from the main array to the cache register, and is busy for tR time. During this time, the GET FEATURE (0Fh) command can be issued to monitor the status of the operation. Following a status of successful completion, the RANDOM DATA READ (03h or 0Bh) command must be issued in order to read the data out of the cache. The RANDOM DATA READ command requires 4 dummy bits, followed by 12-bit column address for the starting byte address. The starting byte address can be 1 to 2176. The Quad Enable bit (QE) of feature (B0[0]) must be set to enable the read from cache x4 command

See **Figure 3.5, 3.6, 3.7, 3.8**.

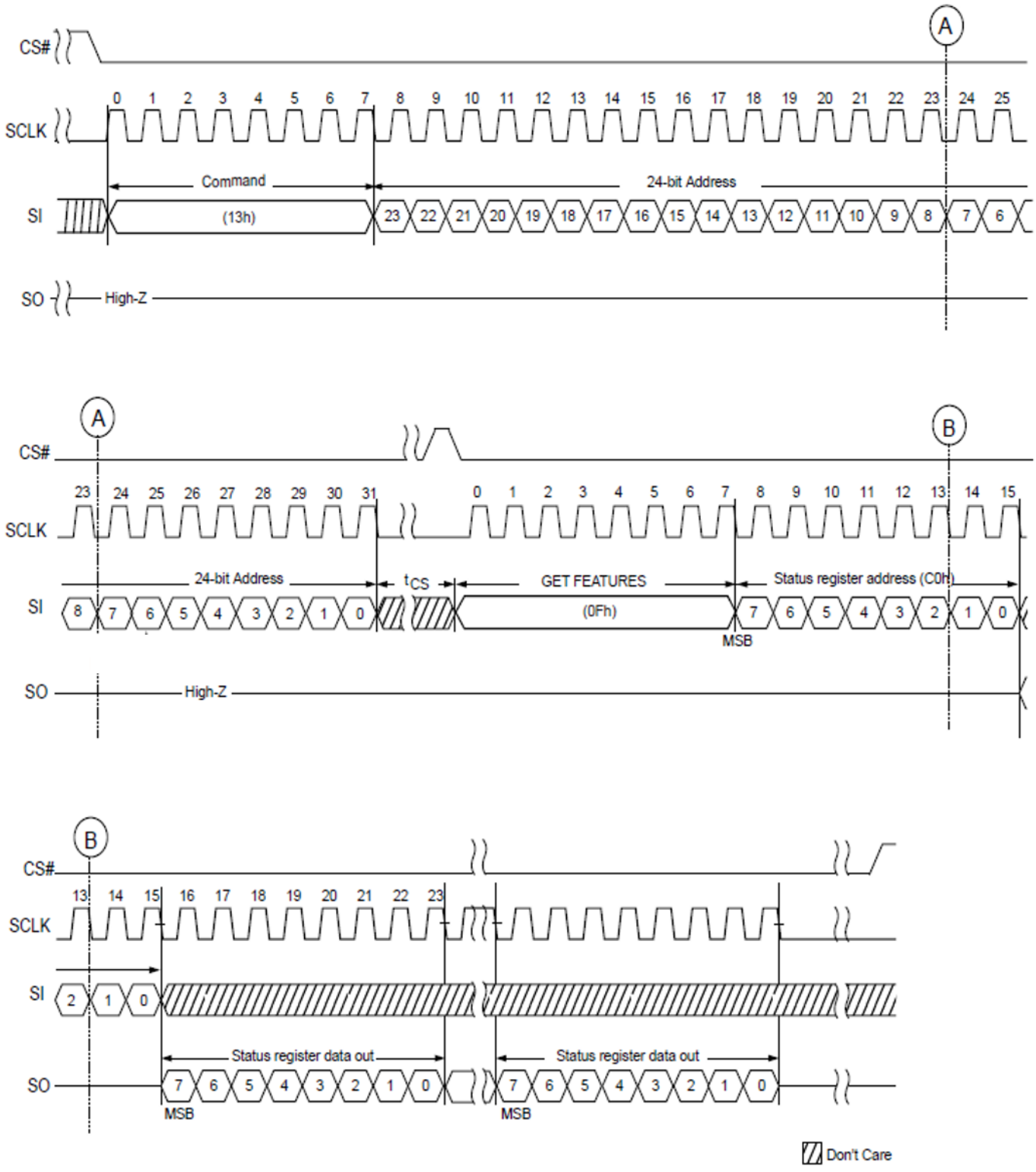



**Figure 3.5** Page Read (13h)

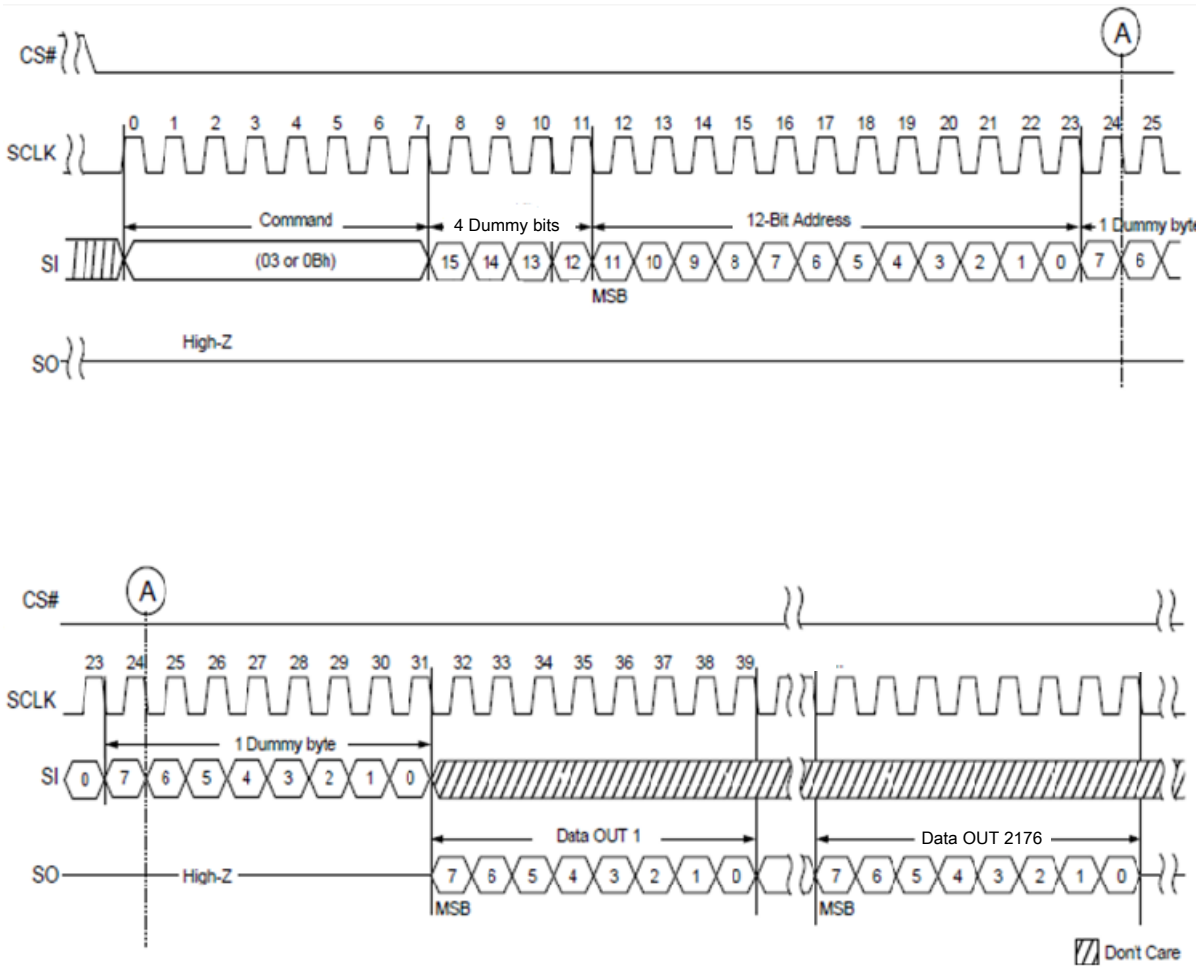



**Figure 3.6** Random Data Read (03h or 0Bh)

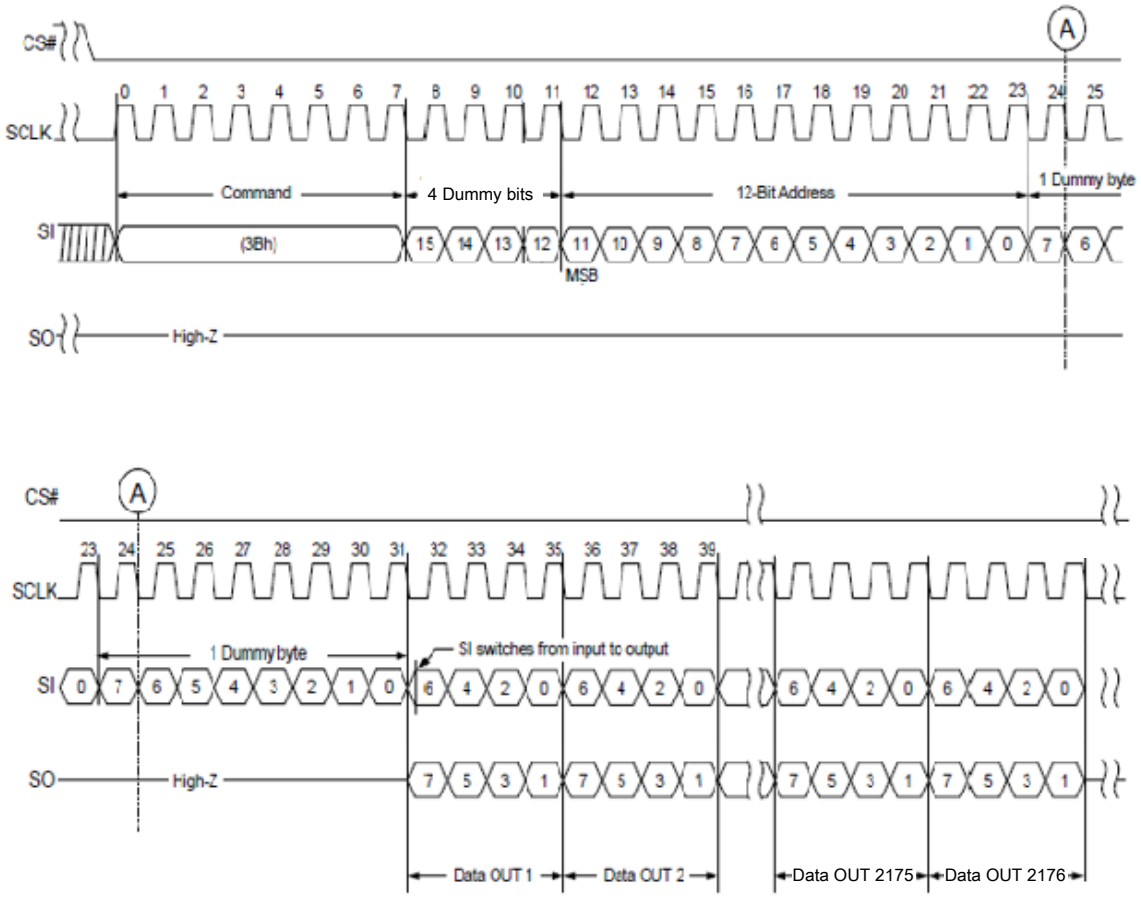



**Figure 3.7** Read From Cache x2

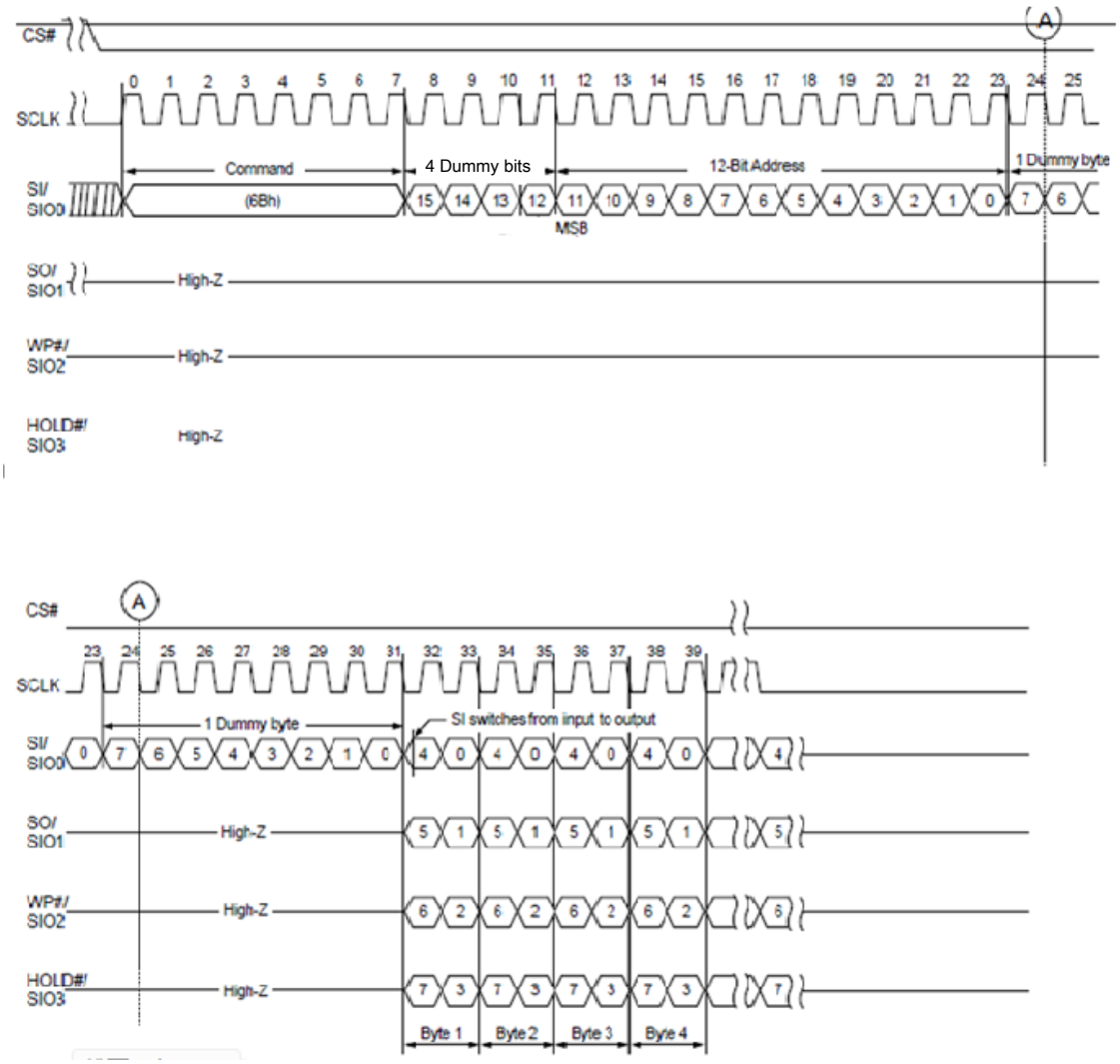



**Figure 3.8** Read From Cache x4

The Quad Enable bit (QE) of feature (B0[0]) must be set to enable the read from cache x4 command

### 3.4 Read ID.

The READ ID command is used to read the 2 bytes of identifier code programmed into the NAND Flash device. The READ ID command reads a 2-byte table (see below) that includes the Manufacturer ID and the device configuration.
See **Figure 3.9**.

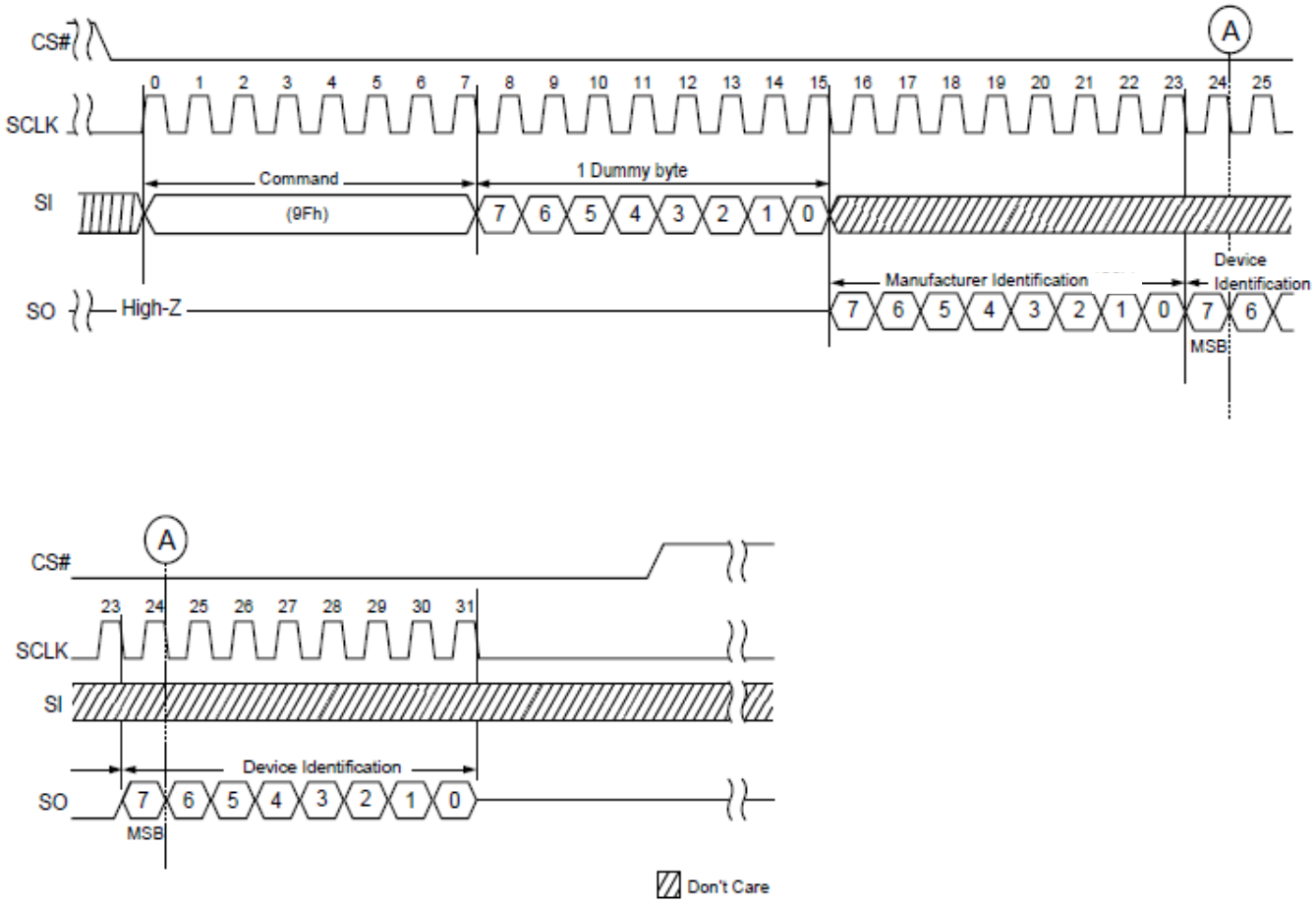



**Figure 3.9** Read ID (9Fh)

| DENSITY | VCC | 1st | 2nd |
|---|---|---|---|
| 4Gbit | 3.3V | E5h | B4h |
| | 1.8V | E5h | 64h |

**Table 3.2** Read ID for Supported Configurations

### 3.5 Parameter Page.

The following command flow must be issued by the memory controller to access the parameter page
1. Issue a SET FEATURES (1Fh) command with a feature address of B0h and data value of 40h (OTP enable, ECC disable).
2. Issue a PAGE READ (13h) command with a block/page address of 0x01h, and then check the status of the read completion using the GET FEATURES (0Fh) command with a feature address of C0h.
3. Issue a READ FROM CACHE (03h) command with an address of 0x00h to read the data out of the NAND device (see the following Parameter Page Data Structure table for a description of the contents of the parameter page.)
4. To exit reading the parameter page, issue a SET FEATURES (1Fh) command with a feature address of B0h and data value of 10h (main array READ, ECC enable).

| Byte | Description | Value |
|---|---|---|
| 0-3 | Parameter page signature | 4Fh, 4Eh, 46h, 49h |
| 4-5 | Revision number | 00h, 00h |
| 6-7 | Features supported | 00h, 00h |
| 8-9 | Optional commands supported | 06h, 00h |
| 10-31 | Reserved (0) | All 00h |
| 32-43 | Device manufacturer (12 ASCII characters) | 44h, 4Fh, 53h, 49h, 4Ch, 49h, 43h, 4Fh, 4Eh, 20h, 20h, 20h |
| 44-63 | Device model (20 ASCII characters) | 3.3V: 44h, 53h, 33h, 35h, 51h, 34h, 47h, 42h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h<br><br>1.8V: 44h, 53h, 33h, 35h, 4Dh, 34h, 47h, 42h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h, 20h |
| 64 | JEDEC manufacturer ID | E5h |
| 65-66 | Date code | 00h, 00h |
| 67-79 | Reserved (0) | All 00h |
| 80-83 | Number of data bytes per page | 00h, 08h, 00h, 00h |
| 84-85 | Number of spare bytes per page | 80h, 00h |
| 86-89 | Number of data bytes per partial page | 00h, 02h, 00h, 00h |
| 90-91 | Number of spare bytes per partial page | 20h, 00h |
| 92-95 | Number of pages per block | 40h, 00h, 00h, 00h |
| 96-99 | Number of blocks per logical unit (LUN) | 00h, 10h, 00h, 00h |
| 100 | Number of logical units (LUNs) | 01h |
| 101 | Number of address cycles (N/A) | 00h |
| 102 | Number of bits per cell | 01h |
| 103-104 | Bad blocks maximum per LUN | 50h, 00h |
| 105-106 | Block endurance | 06h, 04h |
| 107 | Guaranteed valid blocks at beginning of target | 01h |
| 108-109 | Block endurance for guaranteed valid blocks | 01h, 03h |
| 110 | Number of programs per page | 04h |
| 111 | Partial programming attributes | 00h |
| 112 | Number of bits ECC correctability | 08h |

| Byte | Description | Value |
|---|---|---|
| 113 | Number of interleaved address bits | 00h |
| 114 | Interleaved operation attributes | 00h |
| 115-127 | Reserved (0) | All 00h |
| 128 | I/O pin capacitance | 0Ah |
| 129-130 | Timing mode support | 00h, 00h |
| 131-132 | Program cache timing mode support | 00h, 00h |
| 133-134 | $t_{PROG}$ Maximum page program time (µs) | BCh, 02h |
| 135-136 | $t_{BERS}$ Maximum block erase time (µs) | 10h, 27h |
| 137-138 | $t_R$ Maximum page read time (µs) | 3.3V: 78h, 00h<br>1.8V: 82h, 00h |
| 139-163 | Reserved (0) | All 00h |
| 164-165 | Vendor specific Revision number | 00h, 00h |
| 166-253 | Vendor specific | All 00h |
| 254-255 | Integrity CRC | 3.3V:<br>1.8V: |
| 256-511 | Value of bytes 0-255 | |
| 512-767 | Value of bytes 0-255 | |
| 768+ | Additional redundant parameter pages | FFh |

**Table 3.3** Parameter Page Data

### 3.6 UniqueID Page.

The following command flow must be issued by the memory controller to access the uniqueID
1. Issue a SET FEATURES (1Fh) command with a feature address of B0h and data value of 40h (OTP enable, ECC disable).
2. Issue a PAGE READ (13h) command with a block/page address of 0x00h, and then check the status of the read completion using the GET FEATURES (0Fh) command with a feature address of C0h.
3. Issue a READ FROM CACHE (03h) command with an address of 0x00h to read the data out of the NAND device. (The contents of the uniqueID page are described in the following note.)
Note: The device stores 16 copies of the unique ID data. Each copy is 32 bytes; the first 16 bytes are unique data, and the second 16 bytes are the complement of the first 16 bytes. The host should XOR the first 16 bytes with the second 16 bytes. If the result is 16 bytes of FFh, then that copy of the unique ID data is correct. If a non-FFh result is returned, the host can repeat the XOR operation on a subsequent copy of the unique ID data.
4. To exit reading the uniqueID page, issue a SET FEATURES (1Fh) command with a feature address of B0h and data value of 10h (main array READ, ECC enable).

### 3.7 Program Operations.

The PAGE PROGRAM operation sequence programs 1 byte to 2176 bytes of data within a page. The page program sequence is as follows:

- 06h (WRITE ENABLE)
- 02h (PROGRAM LOAD) or 32h (PROGRAM LOAD x4)
- 10h (PROGRAM EXECUTE)
- 0Fh (GET FEATURE command to read the status)

Prior to performing the PROGRAM LOAD operation, a WRITE ENABLE (06h) command must be issued. As with any command that changes the memory contents, the WRITE ENABLE must be executed in order to set the WEL bit. If this command is not issued, then the rest of the program sequence is ignored. WRITE ENABLE must be followed by a PROGRAM LOAD (02h or 32h) command. PROGRAM LOAD consists of an 8-bit Op code, followed by 4 dummy bits and a 12-bit column address, then the data bytes to be programmed. The data bytes are loaded into a cache

register that is 2176 bytes long. Only four partial-page programs are allowed on a single page. If more than 2176 bytes are loaded, then those additional bytes are ignored by the cache register. The command sequence ends when CS goes from LOW to HIGH. **Figure 3.10, 3.11** shows the PROGRAM LOAD operation.

After the data is loaded, a PROGRAM EXECUTE (10h) command must be issued to initiate the transfer of data from the cache register to the main array. PROGRAM EXECUTE consists of an 8-bit Op code, followed by a 24-bit address (6 dummy bits and a 18-bit block/page address). After the page/block address is registered, the memory device starts the transfer from the cache register to the main array, and is busy for tPROG time.

This operation is shown in **Figure 3.12**. During this busy time, the status register can be polled to monitor the status of the operation (refer to the Status Register section). When the operation completes successfully, the next series of data can be loaded with the PROGRAM LOAD command.

See **Figure 3.10, 3.11, 3.12**

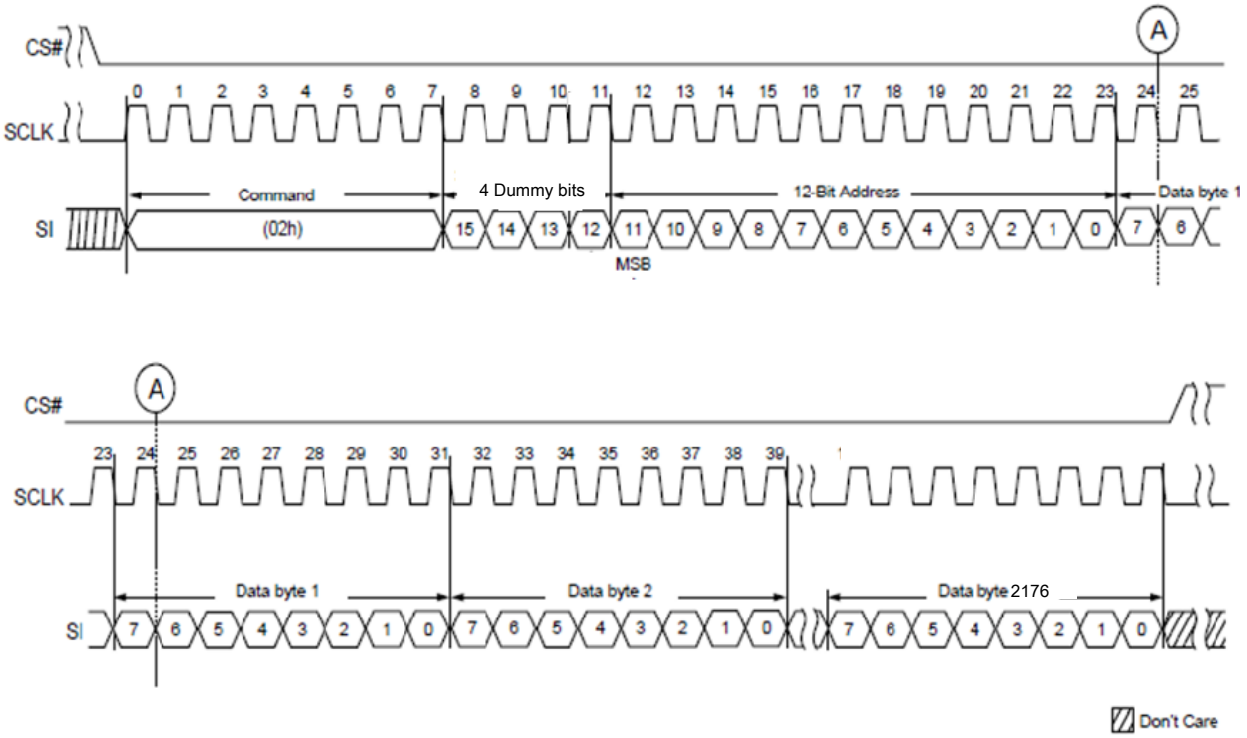



**Figure 3.10** Program Load (02h)

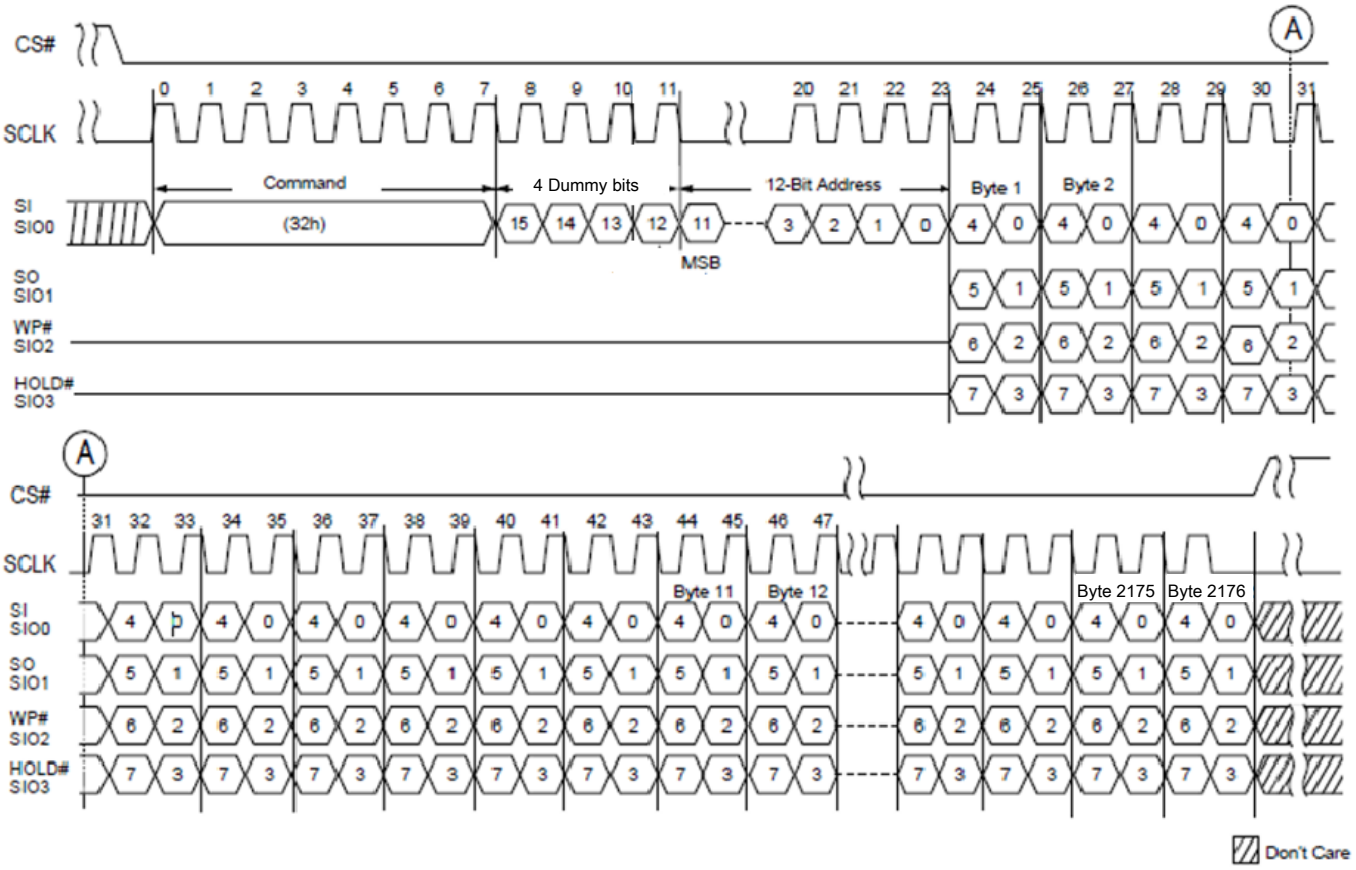



**Figure 3.11** Program Load x4 (32h)

The Quad Enable bit (QE) of feature (B0[0]) must be set to enable the program load x4 command.

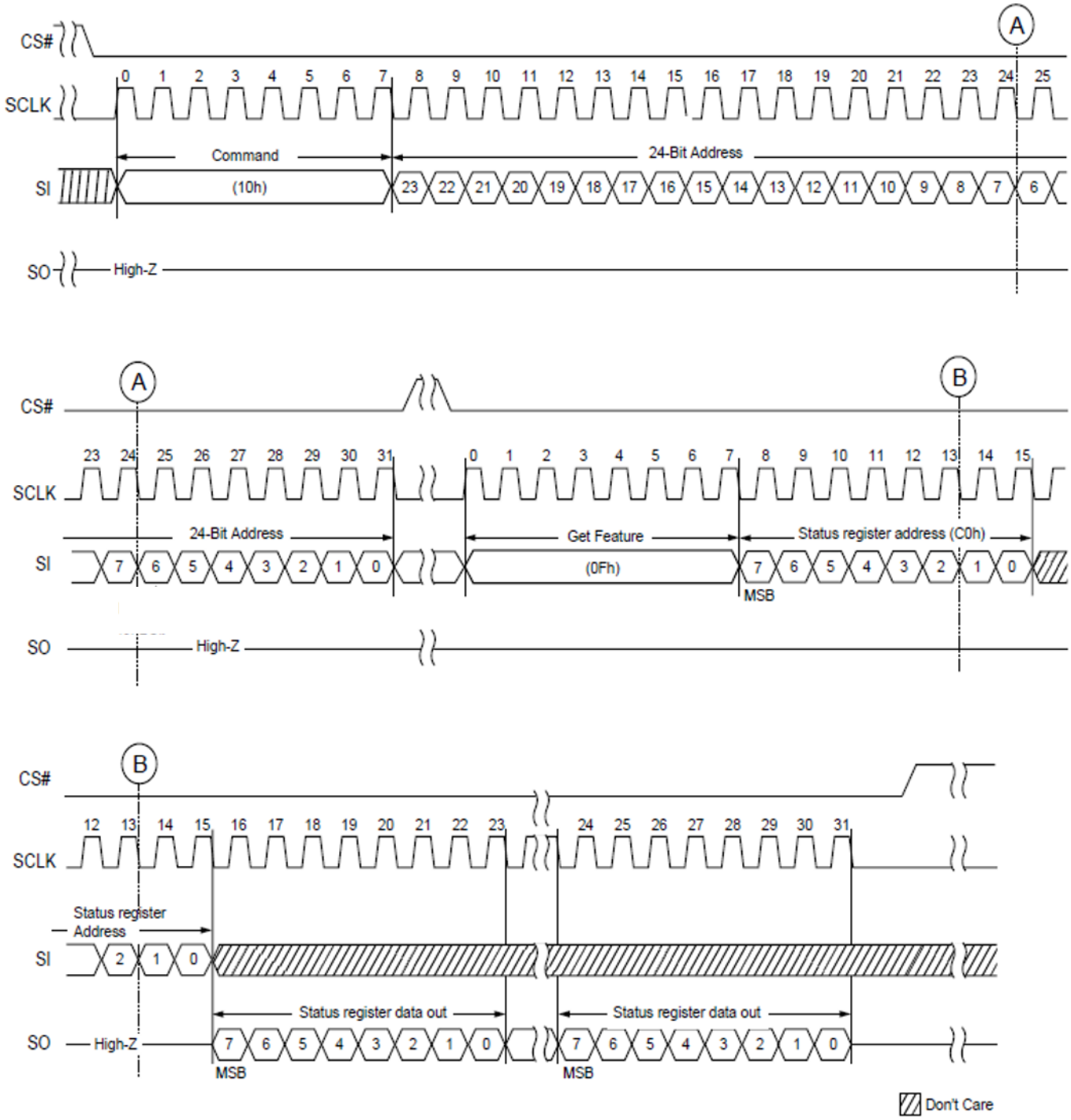



**Figure 3.12** Program Excute (10h)

The RANDOM DATA PROGRAM sequence programs or replaces data in a page with existing data. The random data program sequence is as follows:

- 06h (WRITE ENABLE)
- 84h (PROGRAM LOAD RANDOM DATA) or 34h (PROGRAM LOAD RANDOM DATA x4)
- 10h (PROGRAM EXECUTE)
- 0Fh (GET FEATURE command to read the status)

Prior to performing a PROGRAM LOAD RANDOM DATA operation, a WRITE ENABLE (06h) command must be issued to change the contents of the memory array. Following a WRITE ENABLE (06) command, a PROGRAM LOAD RANDOM DATA (84h) command must be issued. This command consists of an 8-bit Op code, followed by 4 dummy bits and a 12-bit column address. New data is loaded in the column address provided with the 12 bits. If the random data is not sequential, then another PROGRAM LOAD RANDOM DATA (84h) command must be issued with a new column address. After the data is loaded, a PROGRAM EXECUTE (10h) command can be issued to start the programming operation.

See **Figure 3.13, 3.14, 3.12**

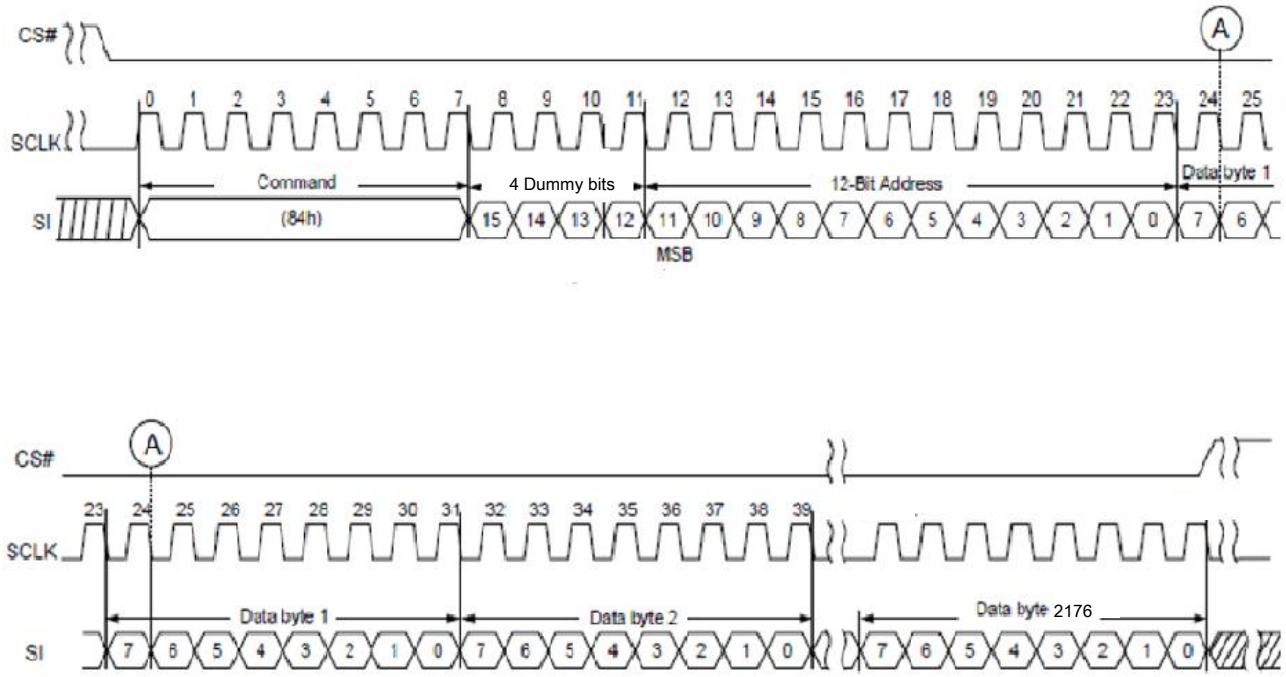



**Figure 3.13** Program Load Random Data (84h)

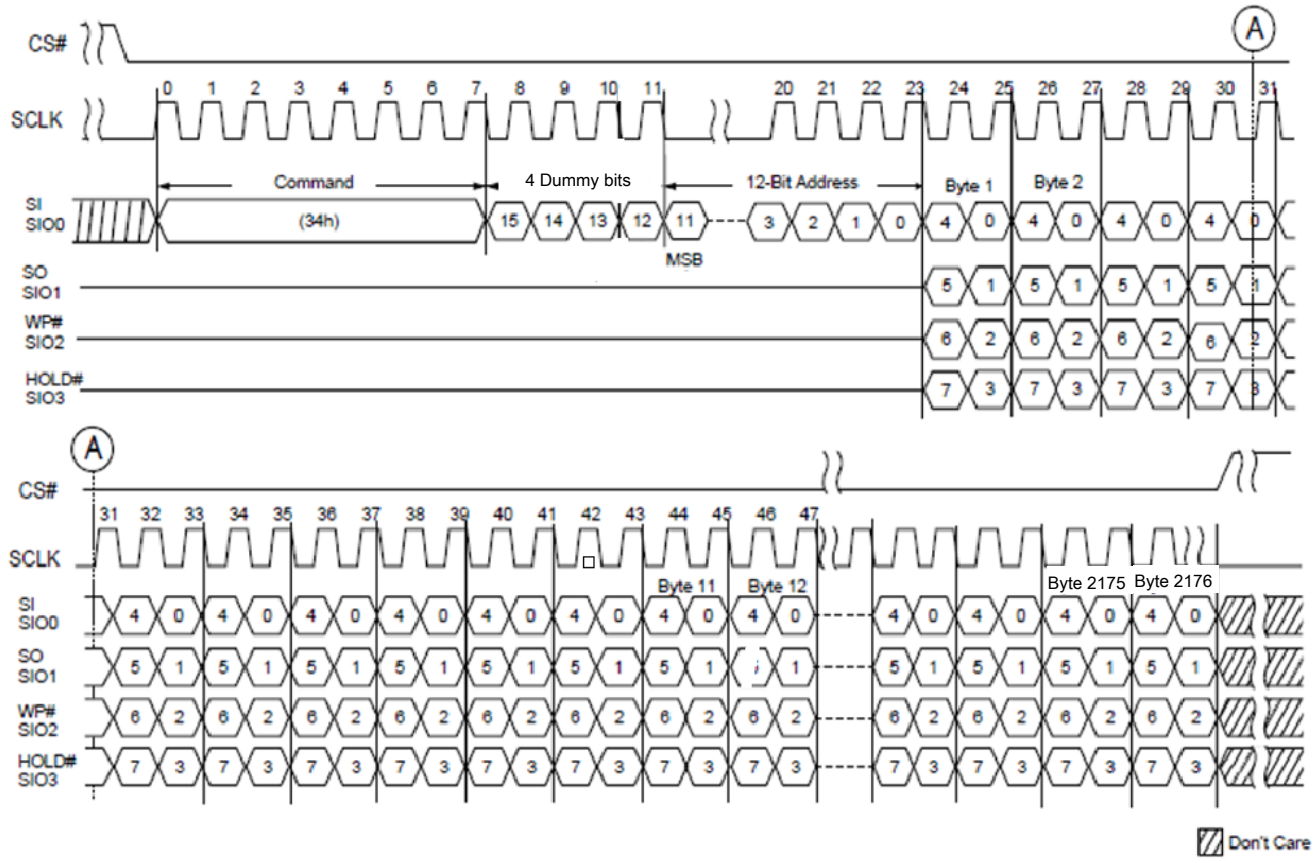



**Figure 3.14** Program Load Random Data x4 (34h)

The INTERNAL DATA MOVE command sequence programs or replaces data in a page with existing data. The INTERNAL DATA MOVE command sequence is as follows:

- 13h (PAGE READ to cache)
- 06h (WRITE ENABLE)
- 84h (PROGRAM LOAD RANDOM DATA) or 34h (PROGRAM LOAD RANDOM DATA x4)
- 10h (PROGRAM EXECUTE)
- 0Fh (GET FEATURE command to read the status)

Prior to performing an internal data move operation, the target page content must be read into the cache register. This is done by issuing a PAGE READ (13h) command. The PAGE READ command must be followed with a WRITE ENABLE (06h) command in order to change the contents of memory array. After the WRITE ENABLE command is issued, the PROGRAM LOAD RANDOM DATA (84h) command can be issued.

This command consists of an 8-bit Op code, followed by 4 dummy bits and a 12-bit column address. New data is loaded in the 12-bit column address. If the random data is not sequential, another PROGRAM LOAD RANDOM DATA (84h) command must be issued with the new column address. After the data is loaded, a PROGRAM EXECUTE (10h) command can be issued to start the programming operation. See **Figure 3.13, 3.14, 3.12**.

### 3.8 BLOCK ERASE

The BLOCK ERASE (D8h) command is used to erase at the block level. The blocks are organized as 64 pages per block, 2176 bytes per page (2048 + 128 bytes). Each block is 136 Kbytes. The BLOCK ERASE command (D8h) operates on one block at a time. The command sequence for the BLOCK ERASE operation is as follows:

- 06h (WRITE ENBALE command)
- D8h (BLOCK ERASE command)
- 0Fh (GET FEATURES command to read the status register)

Prior to performing the BLOCK ERASE operation, a WRITE ENABLE (06h) command must be issued. As with any command that changes the memory contents, the WRITE ENABLE command must be executed in order to set the WEL bit. If the WRITE ENABLE command is not issued, then the rest of the erase sequence is ignored. A WRITE ENABLE command must be followed by a BLOCK ERASE (D8h) command. This command requires a 24-bit address consisting of 6 dummy bits followed by an 18-bit row address.

After the row address is registered, the control logic automatically controls timing and erase-verify operations. The device is busy for tBERS time during the BLOCK ERASE operation.

The GET FEATURES (0Fh) command can be used to monitor the status of the operation.

See **Figure 3.15.**

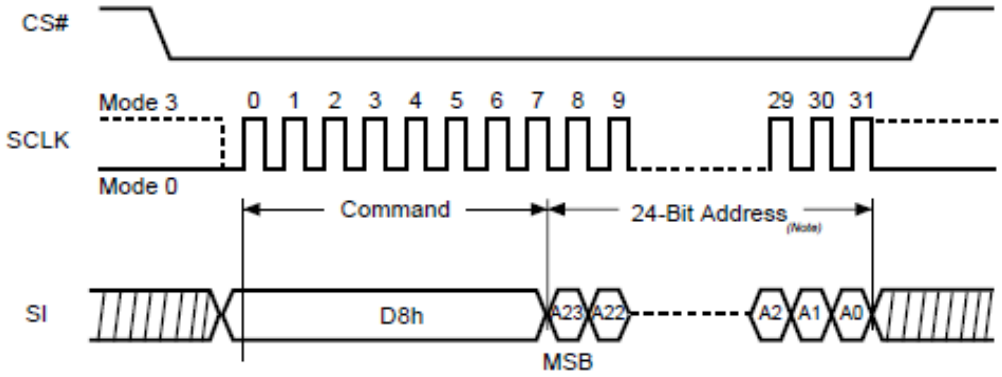



**Figure 3.15** Block Erase

### 3.9 Block Lock Feature.

The block lock feature provides the ability to protect the entire device, or ranges of blocks, from the PROGRAM and ERASE operations. After power-up, the device is in the “locked” state, i.e., bits 1, 2, 3, 4, and 5 of the block lock register are set to 1. To unlock all the blocks, or a range of blocks, the SET FEATURES command must be issued with the A0h feature address, including the data bits shown in Table 7. When BRWD is set and WP is LOW, none of the writable bits (1, 2, 3, 4, 5, and 7) in the block lock register can be set.
When an ERASE command is issued to a locked block, the erase failure, 04h, is returned. When a PROGRAM command is issued to a locked block, program failure, 08h, is returned.

| BP2 | BP1 | BP0 | Invert | Complementary | Protection |
|---|---|---|---|---|---|
| 0 | 0 | 0 | x | x | all unlocked |
| 0 | 0 | 1 | 0 | 0 | upper 1/64 locked |
| 0 | 1 | 0 | 0 | 0 | upper 1/32 locked |
| 0 | 1 | 1 | 0 | 0 | upper 1/16 locked |
| 1 | 0 | 0 | 0 | 0 | upper 1/8 locked |
| 1 | 0 | 1 | 0 | 0 | upper 1/4 locked |
| 1 | 1 | 0 | 0 | 0 | upper 1/2 locked |
| 1 | 1 | 1 | x | x | all locked (default) |
| 0 | 0 | 1 | 1 | 0 | lower 1/64 locked |
| 0 | 1 | 0 | 1 | 0 | lower 1/32 locked |
| 0 | 1 | 1 | 1 | 0 | lower 1/16 locked |
| 1 | 0 | 0 | 1 | 0 | lower 1/8 locked |
| 1 | 0 | 1 | 1 | 0 | lower 1/4 locked |
| 1 | 1 | 0 | 1 | 0 | lower 1/2 locked |
| 0 | 0 | 1 | 0 | 1 | lower 63/64 locked |
| 0 | 1 | 0 | 0 | 1 | lower 31/32 locked |
| 0 | 1 | 1 | 0 | 1 | lower 15/16 locked |
| 1 | 0 | 0 | 0 | 1 | lower 7/8 locked |
| 1 | 0 | 1 | 0 | 1 | lower 3/4 locked |
| 1 | 1 | 0 | 0 | 1 | Block0 |
| 0 | 0 | 1 | 1 | 1 | upper 63/64 locked |
| 0 | 1 | 0 | 1 | 1 | upper 31/32 locked |
| 0 | 1 | 1 | 1 | 1 | upper 15/16 locked |
| 1 | 0 | 0 | 1 | 1 | upper 7/8 locked |
| 1 | 0 | 1 | 1 | 1 | upper 3/4 locked |
| 1 | 1 | 0 | 1 | 1 | Block0 |

**Table 3.4** Definition of Protectiong Bits

### 3.10 OTP Feature.

The serial device offers a protected, one-time programmable NAND Flash memory area. 30 full pages (2176 bytes per page) are available on the device, and the entire range is guaranteed to be good. Customers can use the OTP area any way they want; typical uses include programming serial numbers, or other data, for permanent storage.
To access the OTP feature, the user must issue the SET FEATURES command, followed by feature address B0h. When the OTP is ready for access, pages 02h–1Fh can be programmed in sequential order. The PROGRAM LOAD (02h) and PROGRAM EXECUTE (10h) commands can be used to program the pages. Also, the PAGE READ (13h) command can be used to read the OTP area. The data bits used in feature address B0h to enable OTP access are shown in the table below.

To access OTP, perform the following command sequence:

- Issue the SET FEATURES register write (1Fh)
- Issue the OTP feature address (B0h)
- Issue the PAGE PROGRAM or PAGE READ command

It is important to note that after bits 6 and 7 of the OTP register are set by the user, the OTP area becomes read-only and no further programming is supported. For OTP states, see the following table. If the bit7(OTP_PRT)=1 and bit6(OTP_EN)=0 of B0 register are set by the user, the NAND Flash cannot be programmed.

| OTP Protection Bit | OTP Enabled Bit | State |
|---|---|---|
| 0 | 0 | Normal operation |
| 0 | 1 | Access the Secure OTP |
| 1 | 0 | Not applicable |
| 1 | 1 | OTP Protection by using the Program Execution command (10h) |

**Table 3.5** OTP States

### 3.11 Status Register

The NAND Flash device has an 8-bit status register that software can read during the device operation. The status register will output the status of the operation. The description of data bits from status register are shown in the following table.

| SR Bit | Bit Name | Description |
|---|---|---|
| SR[0] (OIP) | Operation in progress | The bit value indicates whether the device is busy in operations of read/program execute/ erase/ reset command.<br>1: Busy, 0: Ready |
| SR[1] (WEL) | Write enable latch | The bit value indicates whether the device is set to internal write enable latch.<br>When WEL bit sets to 1, which means the internal write enable latch is set, and then the device can accept program/ erase command.<br>1: write enable, 0: not write enable<br>The bit value will be cleared (as "0") by issuing Write Disable command(04h). |
| SR[2] (ERS_Fail) | Erase fail | The bit value shows the status of erase failure or if host erase any invalid address or protected area (including protected blocks or protected Secure OTP area).<br>0: Passed, 1: Failed<br>The bit value will be cleared (as "0") by RESET command or at the beginning of the block erase command operation. |
| SR[3] (PGM_Fail) | Program fail | The bit value shows the status of program failure or if host program any invalid address or protected area (including protected blocks or protected Secure OTP area).<br>0: Passed, 1: Failed<br>The bit value will be cleared (as "0") by RESET command or during the program execute command operation. |
| SR[6:4] (ECC_S2, ECC_S1, ECC_S0) | ECC Status | The bit shows the status of ECC as below:<br>000b = 0 bit error<br>001b = 1 ~ 3 bits error and been corrected.<br>010b = More than 8-bit error and not corrected.<br>011b = 4 ~ 6 bits error and been corrected.<br>101b = 7 ~ 8 bits error and been corrected.<br>others = Reserved<br>The value of ECC_Sx (S2:S0) bits will be clear as "000b" by Reset command or at the start of the Read operation. After a valid Read operation completion, the bit will be updated to reflect the ECC status of the current valid Read operation.<br>The ECC_Sx (S2:S0) value reflects the ECC status of the content of the page 0 of block 0 after a power-on reset.<br>If the internal ECC is disabled by the Set feature command, the ECC_ Sx(S2:S0) are invalid. |
| SR[7] | Reserved | |

**Table 3.6** Status Register Bit Descriptions

### 3.12 ECC Protection

The serial device offers data corruption protection by offering 8-bit internal ECC.
READs and PROGRAMs with internal ECC can be enabled or disabled by setting the ECC bit in the OTP register.
ECC is enabled after device power up, so the default READ and PROGRAM commands operate with internal ECC in the "active" state.
To enable/disable ECC, perform the following command sequence:

- Issue the SET FEATURES register write (1Fh).
- Issue the OTP feature address (B0h).
- Then:
  – To enable ECC Set Bit 4, ECC Enable, to 1.
  – To disable ECC Clear Bit 4, ECC Enable, to 0.

During a PROGRAM operation, the device calculates an ECC code on the 2k page in the cache register, before the page is written to the NAND Flash array. The ECC code is stored in the spare area of the page.
During a READ operation, the page data is read from the array to the cache register, where the ECC code is calculated and compared with the ECC code value read from the array. If a 1- to 8-bit error is detected, the error is corrected in the cache register. Only corrected data is output on the I/O bus. The ECC status bit indicates if the error correction was successful. The ECC Protection table below shows the ECC protection scheme used throughout a page.

With internal ECC, the user must accommodate the following:

- The ECC protection coverage: please refer to Table 3.7. The Distribution of ECC Segment and Spare Area. Only the grey areas are under internal ECC protection when the internal ECC is enabled.
- The number of partial-page program is not 4 in an ECC segment, the user need to program the main area (512B)+spare area(16B) at one program time, so the ECC parity code can be calculated properly and stored in the additional spare area.

| Area | Main Area (0) | Main Area (1) | Main Area (2) | Main Area (3) | Spare Area (0) | Spare Area (1) | Spare Area (2) | Spare Area (3) | Spare Area |
|---|---|---|---|---|---|---|---|---|---|
| Addr. (Start) | 000h | 200h | 400h | 600h | 800h | 810h | 820h | 830h | 840h |
| Addr. (End) | 1FFh | 3FFh | 5FFh | 7FFh | 80Fh | 81Fh | 82Fh | 83Fh | 87Fh |
| Size | 512(B) | 512(B) | 512(B) | 512(B) | 16(B) | 16(B) | 16(B) | 16(B) | 64(B) |

**Table 3.7** The Distribution of ECC Segment and Spare Area

*800h : reserved for bad block data.*
*840h ~ 87Fh : Internal ECC parity data. It will be overwritten by ECC parity generation.*
*Grey area: Under ECC protection*

### 3.13 Block Protection (One Time Program)

Some range of blocks can be protected by using block protection command. It means that program and erase operations can not be executed in protected blocks. 0~255, 0~511, 0~1023, 0~2047 blocks can be protected by each block protection command(B1h, B2h, B3h, B4h). Because block protection is nonvolatile, a power-on or power-off sequence does not affect the block status after the block protect command is issued.

The operation sequence is as below

- 06h (WRITE ENABLE)
- B1h ~B4h (PERMANENT BLOCK LOCK PROTECTION)
- 24-bit address (it should be all zero)
- After tPROG time, use GET FEATURE command (0Fh) with feature address C0h

| Proteced Block Command | B1h | B2h | B3h | B4h |
|---|---|---|---|---|
| Protected Block Range | 0~255 | 0~511 | 0~1023 | 0~2047 |

**Table 3.8** The Permanent block protection table

# 4 Device Parameters

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Valid Block Number, 4Gb | $N_{VB}$ | 4016 | | 4096 | Blocks |

**Table 4.1** Valid Blocks Number

The First block (Block 0) is guaranteed to be a valid block at the time of shipment.
The specification for the minimum number of valid blocks is applicable over lifetime.

| Symbol | Parameter | Value | | Unit |
|---|---|---|---|---|
| | | 1.8V | 3.3V | |
| $T_A$ | Ambient Operating Temperature | –40 to 85 | –40 to 85 | °C |
| | | –40 to 105 | –40 to 105 | |
| | | –40 to 125 | –40 to 125 | |
| $T_{BIAS}$ | Temperature Under Bias | –50 to 125 | –50 to 125 | °C |
| $T_{STG}$ | Storage Temperature | –65 to 150 | –65 to 150 | °C |
| $V_{IO}$ | Input or Output Voltage | –0.6 to 2.7 | –0.6 to 4.6 | V |
| $V_{CC}$ | Supply Voltage | –0.6 to 2.7 | –0.6 to 4.6 | V |

**Table 4.2** Absolute Maximum Ratings

.

| Parameter | | Symbol | Test Conditions | 1.8Volt | | | 3.3Volt | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Typ | Max | Min | Typ | Max | |
| Operating Current | Sequential Read | $I_{CC1}$ | 83/104Mhz, CS#=$V_{IL}$, $I_{OUT}$=0mA | - | 10 | 20 | - | 15 | 30 | mA |
| | Program | $I_{CC2}$ | - | - | 10 | 20 | - | 15 | 30 | mA |
| | Erase | $I_{CC3}$ | - | - | 10 | 20 | - | 15 | 30 | mA |
| Stand-by Current (TTL) | | $I_{CC4}$ | CS#= $V_{CC}$, VIN=VCC or GND | - | - | 1 | | | 1 | mA |
| Stand-By Current (CMOS) | | $I_{CC5}$ | CS#=$V_{CC}$, VIN=VCC or GND | - | 10 | 50 | | 10 | 50 | uA |
| Input Leakage Current | | $I_{LI}$ | $V_{IN}$=0 to Vcc (max) | - | - | ±10 | | - | ±10 | uA |
| Output Leakage Current | | $I_{LO}$ | $V_{OUT}$=0 to Vcc(max) | - | - | ±10 | | - | ±10 | uA |
| Input High Voltage | | $V_{IH}$ | - | 0.8x$V_{CC}$ | - | $V_{CC}$ +0.3 | 0.8x $V_{CC}$ | - | $V_{CC}$ +0.3 | V |
| Input Low Voltage | | $V_{IL}$ | - | -0.3 | - | 0.2x $V_{CC}$ | -0.3 | - | 0.2x$V_{CC}$ | V |
| Output High Voltage Level | | $V_{OH}$ | $I_{OH}$ = -100uA | $V_{CC}$-0.1 | - | - | | | | V |
| | | | $I_{OH}$ = -400uA | | | | 2.4 | - | - | V |
| Output Low Voltage Level | | $V_{OL}$ | $I_{OL}$ = 100uA | - | - | 0.1 | | | | V |
| | | | $I_{OL}$ = 2.1mA | | | | - | - | 0.4 | V |

**Table 4.3** DC and Operating Characteristics

| Parameter | Value | |
|---|---|---|
| | 1.8Volt | 3.3Volt |
| Input Pulse Levels | 0V to $V_{CC}$ | 0V to $V_{CC}$ |
| Input Rise and Fall Times | 5ns | 5ns |
| Input and Output Timing Levels | $V_{CC}$ / 2 | $V_{CC}$ / 2 |
| Output Load (1.7V – 1.95V & 2.5V - 3.6V) | 1 TTL GATE and CL=30pF | 1 TTL GATE and CL=30pF |

**Table 4.4** AC Test Conditions

| Item | Symbol | Test Condition | Min | Max | Unit |
|---|---|---|---|---|---|
| Input / Output Capacitance (1) | $C_{I/O}$ | $V_{IL}$ = 0V | - | 10 | pF |
| Input Capacitance (1) | $C_{IN}$ | $V_{IN}$ = 0V | - | 10 | pF |

**Table 4.5** Ping Capacitance (TA=25C,f=1.0MHz)

| Parameter | | Symbol | Min | | Typ | | Max | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1.8Volt | 3.3Volt | 1.8Volt | 3.3Volt | 1.8Volt | 3.3Volt | |
| Read Time | | $t_R$ | - | - | - | - | 25 | 25 | us |
| Read Time with internal ECC enabled | | $t_{R_ECC}$ | 80 | 70 | - | - | 130 | 120 | us |
| Program Time | | $t_{PROG}$ | - | - | 300 | 300 | 700 | 700 | us |
| Program Time with internal ECC enabled | | $t_{PROG_ECC}$ | - | - | 320 | 320 | 700 | 700 | us |
| Number of partial Program Cycles in the same page | Main + Spare Array | NOP | - | - | - | - | 4 | 4 | Cycle |
| Block Erase Time | | $t_{BERS}$ | - | - | 2.0 | 2.0 | 10 | 10 | ms |

**Table 4.6** Read/Program/Erase Characteristics

| Parameter | Symbol | 1.8 Volt | | 3.3 Volt | | Unit |
|---|---|---|---|---|---|---|
| | | Min | Max | Min | Max | |
| Clock Frequency | $f_C$ | | 83 | | 104 | MHz |
| Clock HIGH time | $t_{WH}$ | 4 | | 4 | | ns |
| Clock LOW time | $t_{WL}$ | 4 | | 4 | | ns |
| Command deselect time | $t_{CS}$ | 100 | | 100 | | ns |
| Chip select# hold time | $t_{CHSH}$ | 5 | | 5 | | ns |
| Chip select# setup time | $t_{SLCH}$ | 5 | | 5 | | ns |
| Chip select# non-active setup time | $t_{SHCH}$ | 5 | | 5 | | ns |
| Chip select# non-active hold time | $t_{CHSL}$ | 5 | | 5 | | ns |
| Output disable time | $t_{DIS}$ | | 20 | | 20 | ns |
| Hold# non-active setup time relative to SCLK | $t_{HC}$ | 5 | | 5 | | ns |
| Hold# setup time relative to SCLK | $t_{HD}$ | 5 | | 5 | | ns |
| Data input hold time | $t_{HDDAT}$ | 3.5 | | 3.5 | | ns |
| Output hold time | $t_{HO}$ | 1 | | 1 | | ns |
| Hold to output High-Z | $t_{HZ}$ | | 15 | | 15 | ns |
| Hold to output low-Z | $t_{LZ}$ | | 15 | | 15 | ns |
| Data input setup time | $t_{SUDAT}$ | 3.5 | | 3.5 | | ns |
| Clock LOW to output Valid | $t_V$ | | 10 | | 8 | ns |
| WP# hold time | $t_{WPH}$ | 100 | | 100 | | ns |
| WP# setup time | $t_{WPS}$ | 20 | | 20 | | ns |
| HOLD# high hold time relative to SCLK | $t_{CHHH}$ | 5 | | 5 | | ns |
| HOLD# low hold time relative to SCLK | $t_{CHHL}$ | 5 | | 5 | | ns |
| Device Resetting Time (Read/Program/Erase) | $t_{RST}$ | | 5/10/500 (1) | | 5/10/500 (1) | us |

**Table 4.7** AC Timing Characteristics

**NOTE:**

(1) If Reset Command (FFh) is written at Ready state, the device goes into Busy for maximum 5us

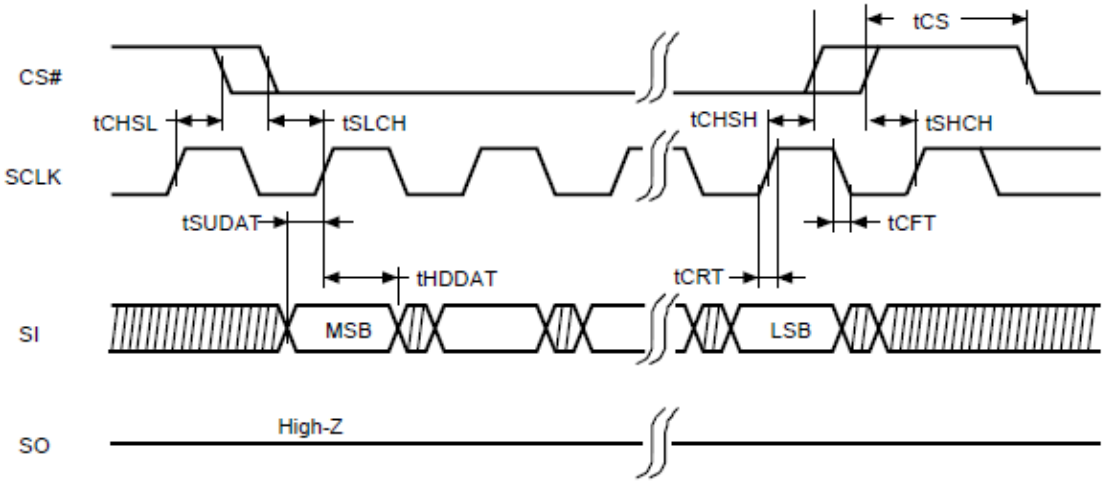



**Figure 4.1** Serial Input Timing

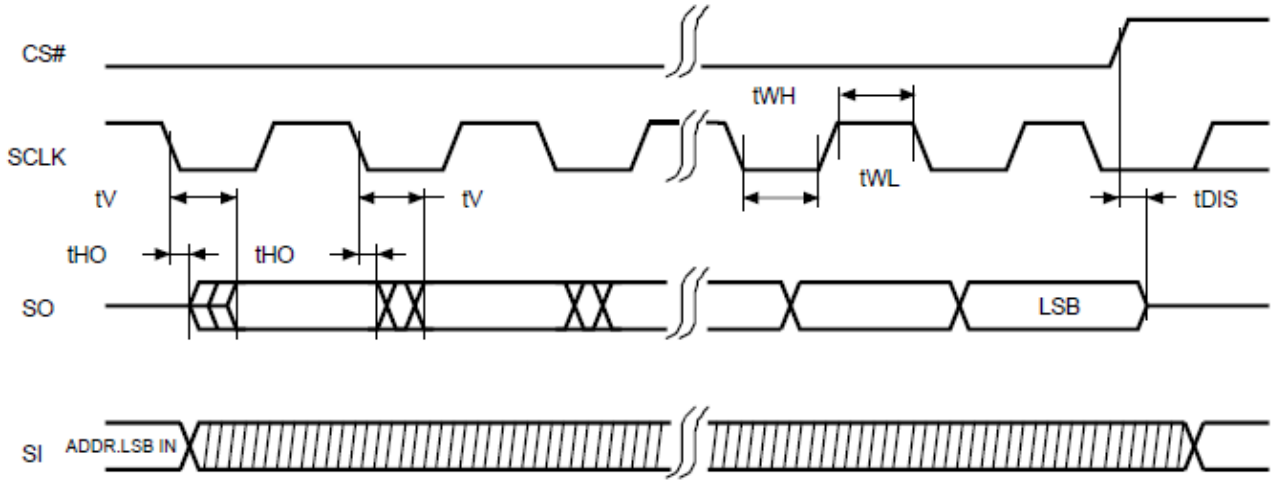



**Figure 4.2** Serial Output Timing

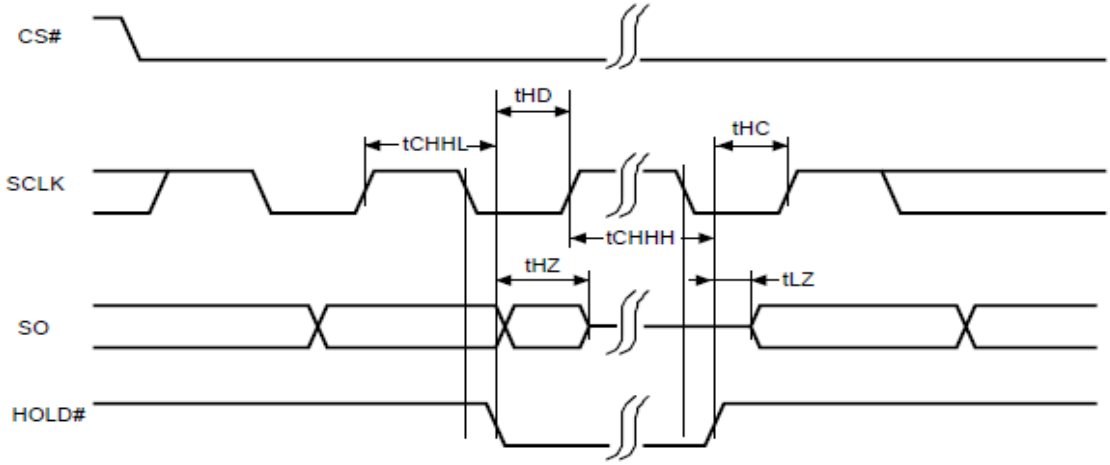



**Figure 4.3** Hold Timing

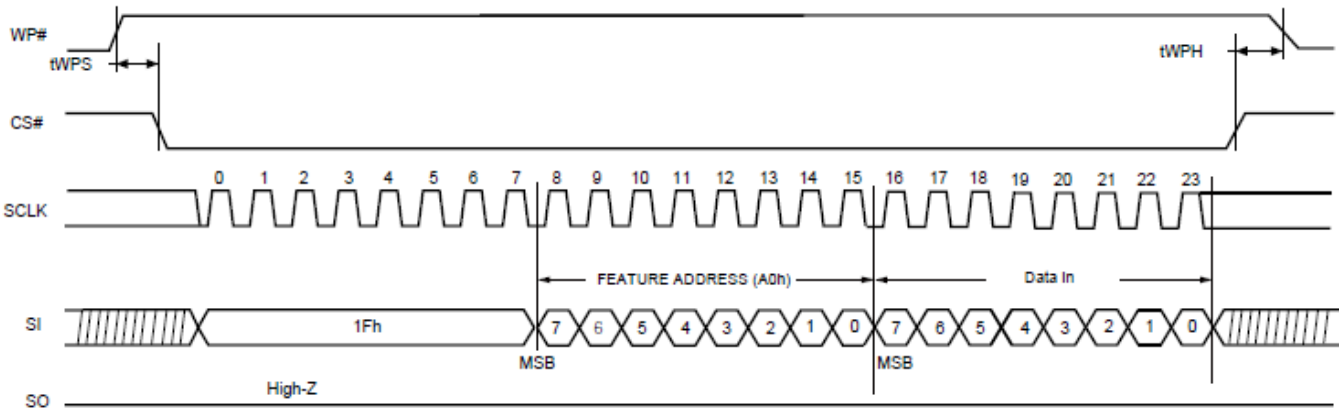



**Figure 4.4** WP# Setup/Hold Timing When BPRWD=1

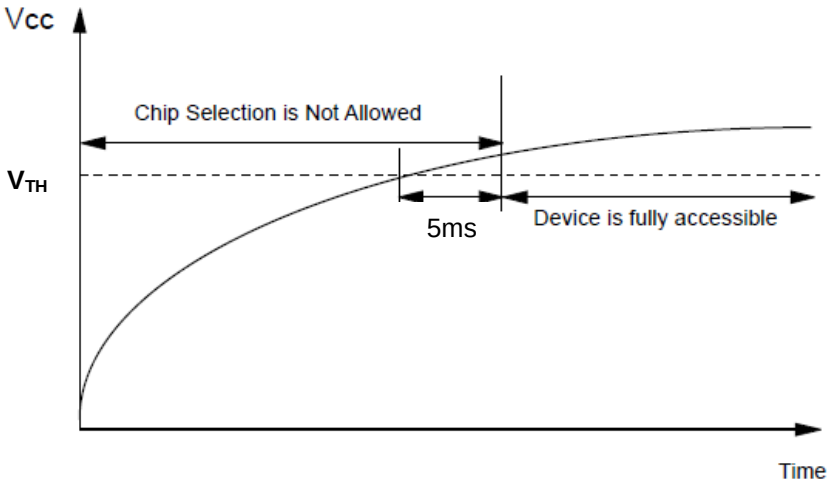



**Figure 4.5** Power on Sequence

**Note :** $V_{TH}$ = 1.5 Volt for 1.8 Volt Supply devices; 2.5 Volt for 3.3 Volt Supply devices

# 5 Bad Block Management

Devices with Bad Blocks have the same quality level and the same AC and DC characteristics as devices where all the blocks are valid. A Bad Block does not affect the performance of valid blocks because it is isolated from the bit line and common source line by a select transistor. The devices are supplied with all the locations inside valid blocks erased (FFh). The Bad Block Information is written prior to shipping. Any block where the 1st Byte in the spare area of the 1st or 2nd page (if the 1st page is Bad) does not contain FFh is a Bad Block. The Bad Block Information must be read before any erase is attempted as the Bad Block Information may be erased. For the system to be able to recognize the Bad Blocks based on the original information it is recommended to create a Bad Block table following the flowchart

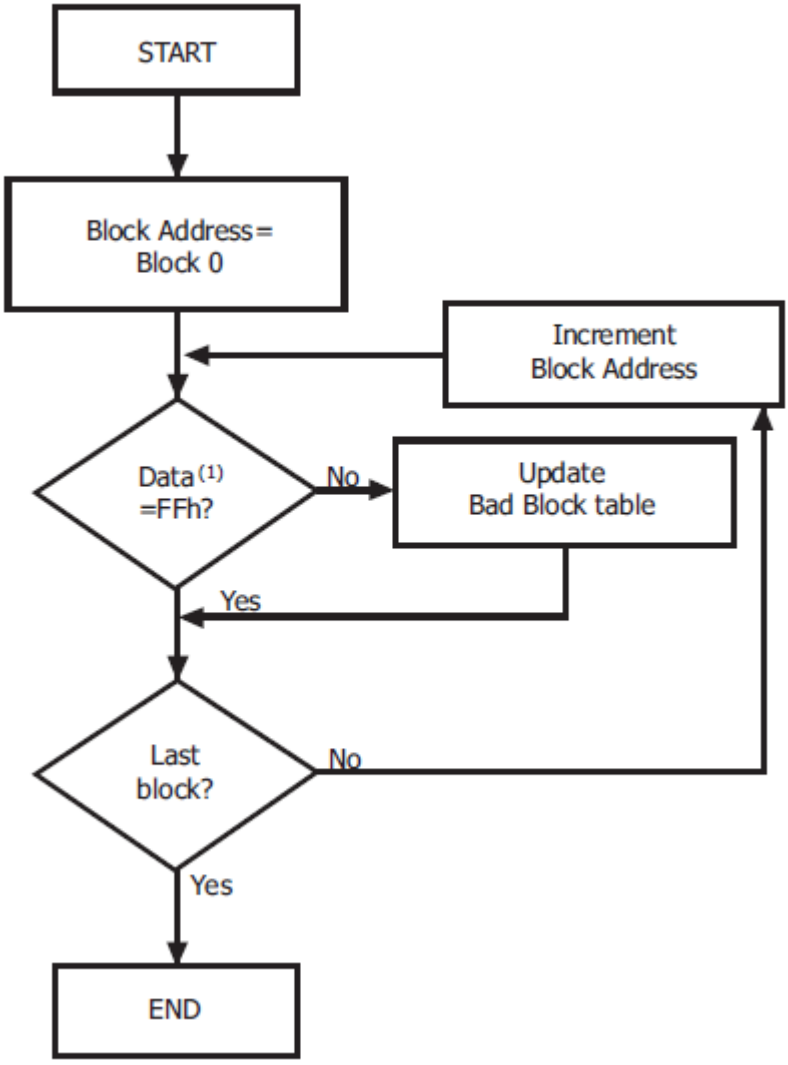



**Figure 5.1** Bad Block Management Flowchart

Over the lifetime of the device additional Bad Blocks may develop. In this case the block has to be replaced by copying the data to a valid block. These additional Bad Blocks can be identified as attempts to program or erase them will give errors in the Status Register.
The failure of a page program operation does not affect the data in other pages in the same block, the block can be replaced by re-programming the current data and copying the rest of the replaced block to an available valid block.

| Failure Mode | | Sequence |
|---|---|---|
| Write | Erase Failure | Block Replacement |
| | Program Failure | Block Replacement |
| Read | Read Failure | ECC |

**Table 5.1** Block Failure

Block Replacement flow is as below
1.When an error happens in the nth page of the Block 'A' during erase or program operation.
2.Copy the data in the 1st ~ (n-1)th page to the same location of another free block. (Block 'B')
3.Copy the nth page data of the Block 'A' in the buffer memory to the nth page of the Block 'B'.
4.Do not erase or program to Block 'A' by creating an 'invalid block' table or other appropriate scheme.

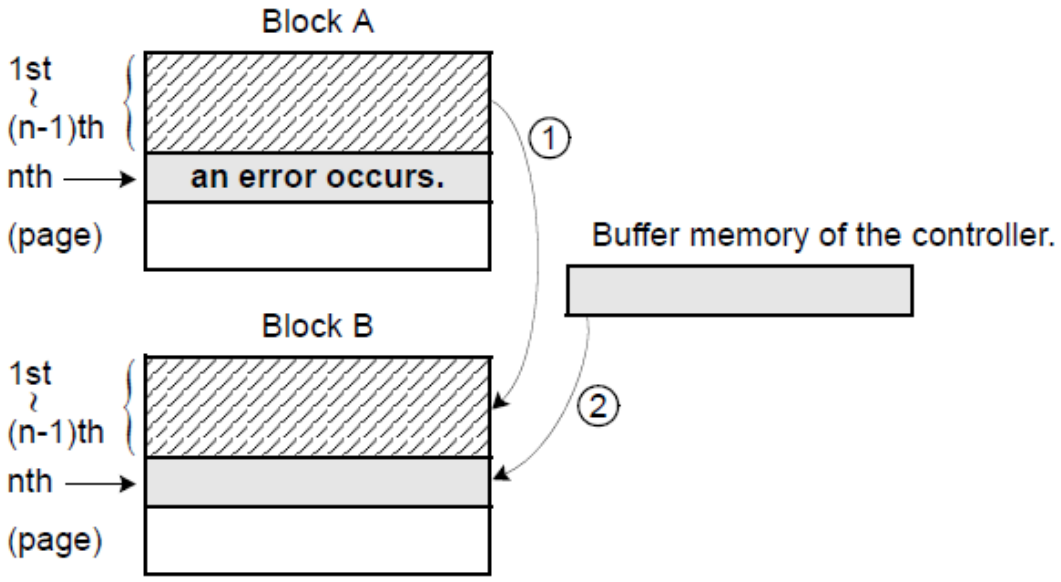
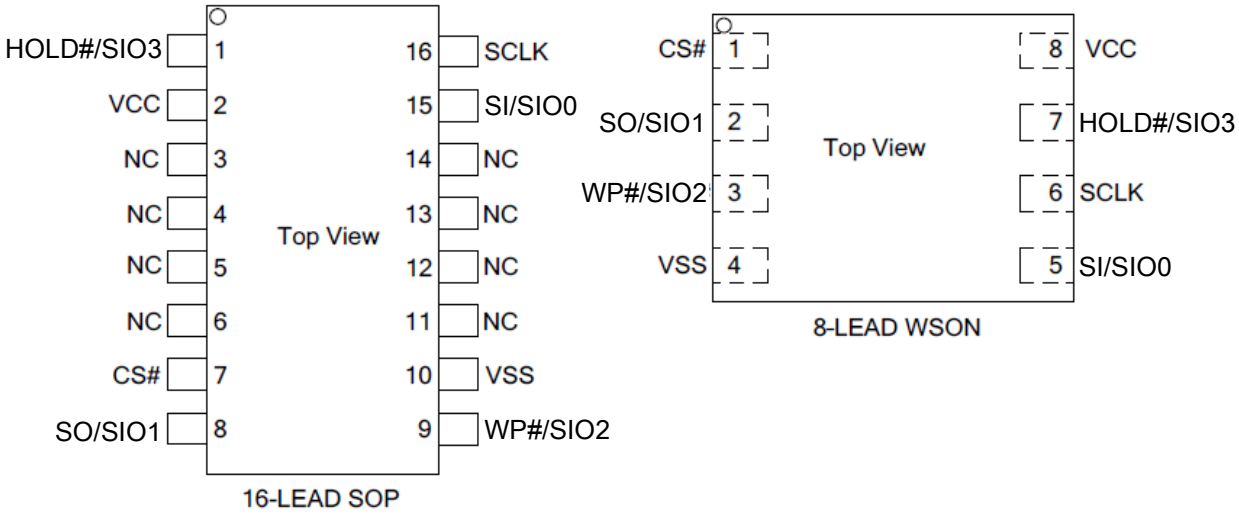


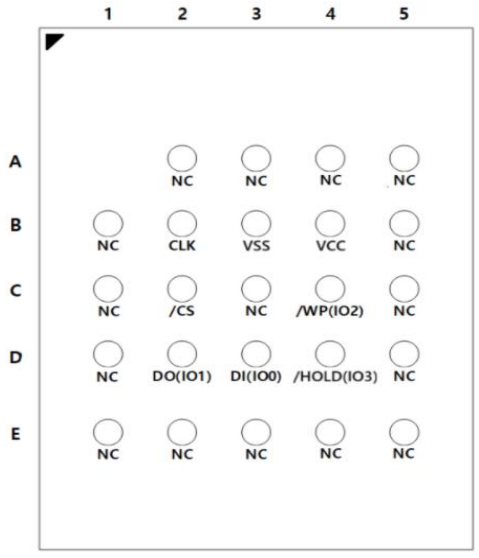

**Figure 5.2** Bad Block Replacement

.

# 6 Supported Packages

## 6.1 PIN CONFIGURATION



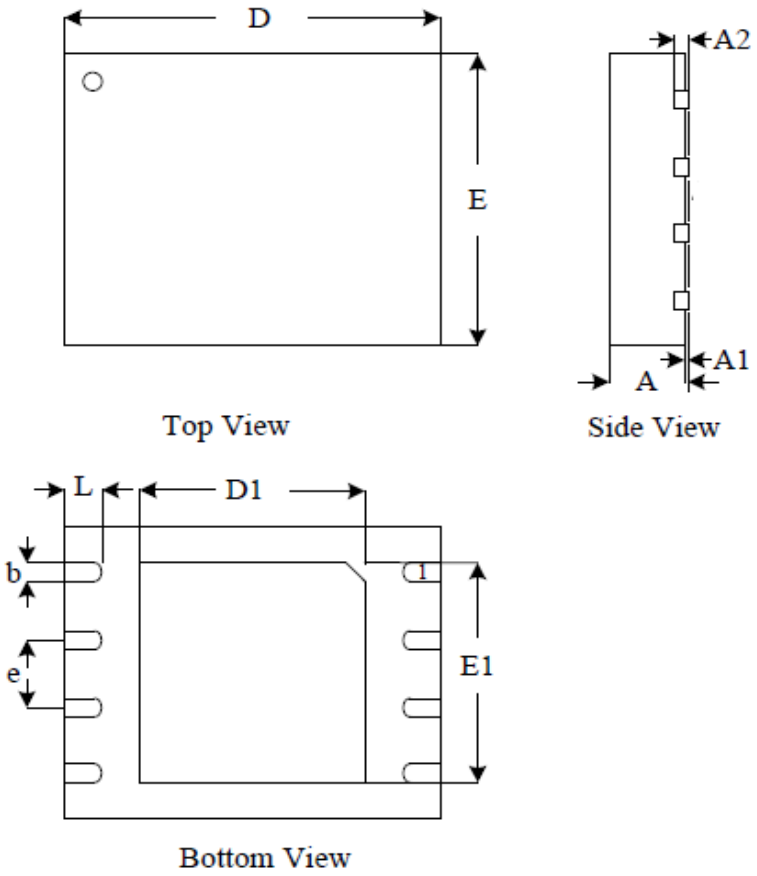




BGA24(5x5 Ball Array)

**Figure 6.1** Pin Configuration

## 6.2 PACKAGE DIMENSIONS

**WSON(8*6mm)**

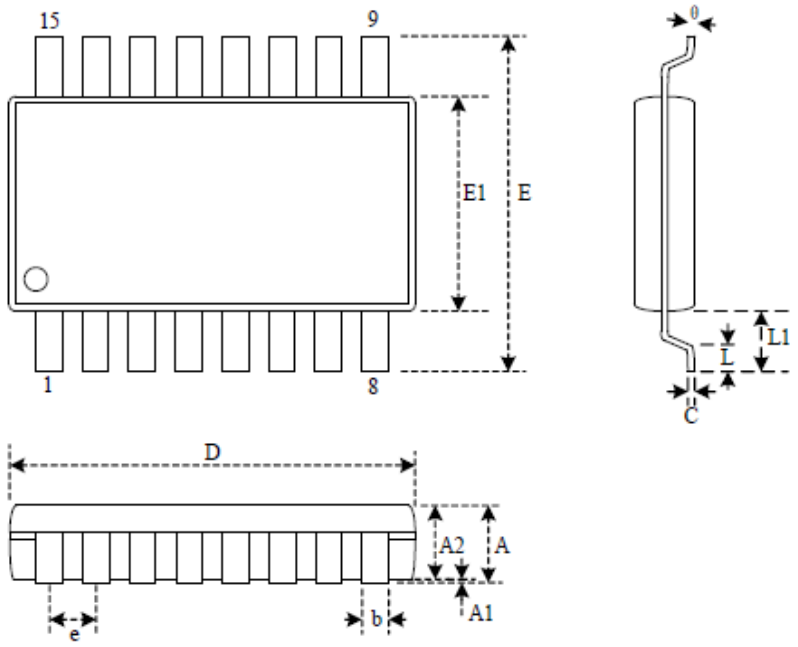



**Dimensions**

| Symbol / Unit | | A | A1 | A2 | b | D | D1 | E | E1 | e | L |
|---|---|---|---|---|---|---|---|---|---|---|---|
| mm | Min | 0.70 | | | 0.35 | 7.95 | 3.25 | 5.95 | 4.15 | | 0.40 |
| | Nom | 0.75 | | 0.20 | 0.40 | 8.00 | 3.40 | 6.00 | 4.30 | 1.27 | 0.50 |
| | Max | 0.80 | 0.05 | | 0.45 | 8.05 | 3.50 | 6.05 | 4.40 | | 0.60 |
| Inch | Min | 0.028 | | | 0.014 | 0.313 | 0.128 | 0.234 | 0.163 | | 0.016 |
| | Nom | 0.030 | | 0.008 | 0.016 | 0.315 | 0.134 | 0.236 | 0.169 | 0.05 | 0.020 |
| | Max | 0.032 | 0.002 | | 0.019 | 0.317 | 0.138 | 0.238 | 0.173 | | 0.024 |

**Figure 6.2** WSON (8*6mm)

**SOP16**

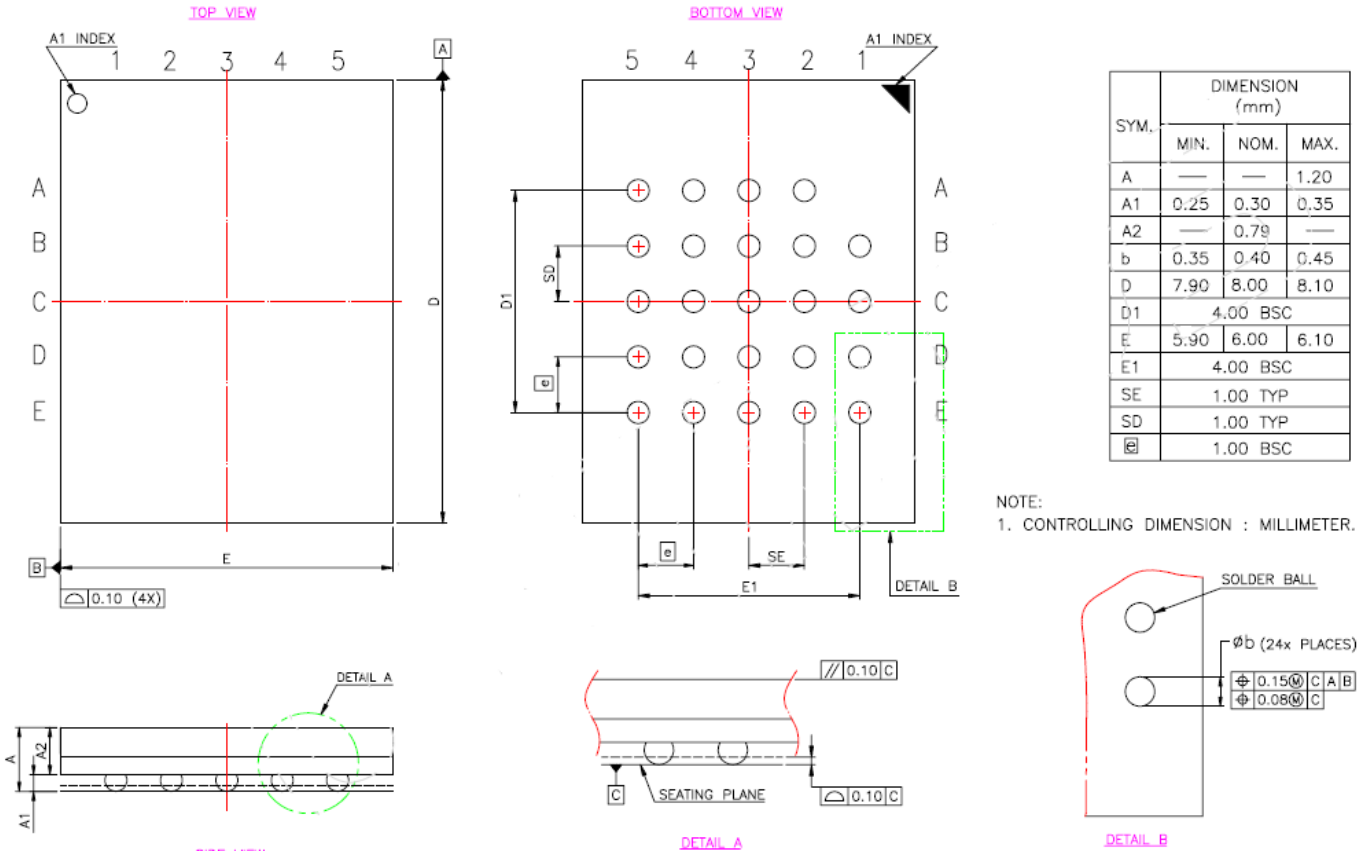



**Dimensions**

| Symbol / Unit | | A | A1 | A2 | b | C | D | E | E1 | e | L | L1 | S | θ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| mm | Min | 2.36 | 0.10 | 2.24 | 0.36 | 0.20 | 10.10 | 10.10 | 7.42 | | 0.40 | 1.31 | 0.51 | 0 |
| | Nom | 2.55 | 0.20 | 2.34 | 0.41 | 0.25 | 10.30 | 10.35 | 7.52 | 1.27 | 0.84 | 1.44 | 0.64 | 5 |
| | Max | 2.75 | 0.30 | 2.44 | 0.51 | 0.30 | 10.50 | 10.60 | 7.60 | | 1.27 | 1.57 | 0.77 | 8 |
| Inch | Min | 0.093 | 0.004 | 0.088 | 0.014 | 0.008 | 0.397 | 0.397 | 0.292 | | 0.016 | 0.052 | 0.020 | 0 |
| | Nom | 0.100 | 0.008 | 0.092 | 0.016 | 0.010 | 0.405 | 0.407 | 0.296 | 0.050 | 0.033 | 0.057 | 0.025 | 5 |
| | Max | 0.108 | 0.012 | 0.096 | 0.020 | 0.012 | 0.413 | 0.417 | 0.299 | | 0.050 | 0.062 | 0.030 | 8 |

**Figure 6.3** SOP16

**BGA24(5x5 Ball Array)**




| SYM. | DIMENSION (mm) | | |
|---|---|---|---|
| | MIN. | NOM. | MAX. |
| A | — | — | 1.20 |
| A1 | 0.25 | 0.30 | 0.35 |
| A2 | — | 0.79 | — |
| b | 0.35 | 0.40 | 0.45 |
| D | 7.90 | 8.00 | 8.10 |
| D1 | 4.00 BSC | | |
| E | 5.90 | 6.00 | 6.10 |
| E1 | 4.00 BSC | | |
| SE | 1.00 TYP | | |
| SD | 1.00 TYP | | |
| e | 1.00 BSC | | |

NOTE:
1. CONTROLLING DIMENSION : MILLIMETER.

**Figure 6.4** BGA24(5x5 Ball Array)